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Strain-coupled one-dimensional turbulence for rapid distortion

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The distortion of turbulence by a mean strain—amplifying some components, redistributing energy through the fluctuating pressure, and rescaling the spectrum—is central to flows from wind-tunnel contractions to the stagnation regions of lifting surfaces. Capturing it economically is difficult: scale-resolving simulation is costly, linear rapid-distortion theory omits the nonlinear relaxation acting over finite strain, and second-moment closures discard the spectral information the distortion redistributes. We present a strain-coupled formulation of one-dimensional turbulence (ODT) that evolves strained turbulence with full scale resolution at negligible cost. The mean-strain production is imposed as a continuous forcing of the line velocity, shown to preserve the structural invariants of the triplet-map mechanism; the rapid pressure–strain enters as an energy-conserving redistribution operator constructed to reproduce the homogeneous rapid-distortion evolution of the component energies exactly under an explicit spectral closure; and the straining of scales is carried by a dilatation of the domain. Two results follow. First, the model produces a broadband distorted spectrum departing measurably from the rigid translation linear theory predicts—a difference of spectral shape no linearly strained input can represent—at a strain-to-turbulence ratio $\chi = Sk_t/\varepsilon \approx 0.8$, the intermediate regime the applications occupy. Second, the rapid pressure–strain term, exactly a nonlocal three-dimensional spectral integral, collapses to a closed single-line functional under one physically grounded assumption, axisymmetry of the undistorted turbulence about the line. The formulation reproduces rapid-distortion theory exactly at onset and the anisotropy of the direct numerical simulations of Lee & Reynolds (1985) quantitatively; external validation of the distorted spectrum, and its coupling to airfoil–turbulence interaction noise, are deferred to a companion paper.

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1. Introduction

The distortion of turbulence by a mean velocity gradient is among the most basic and far-reaching problems in turbulence dynamics. When a turbulent flow is strained—accelerated through a contraction, deflected around a bluff body, sheared in a boundary layer, or compressed across a shock—the mean deformation reorganises the fluctuating field: it amplifies some velocity components and suppresses others, redistributes energy among them through the fluctuating pressure, and rescales the spectral content of the motion. When the strain is applied over a time short compared with the eddy-turnover time of the energy-containing scales, the nonlinear interactions among eddies cannot keep pace and the distortion is governed, to leading order, by the linear dynamics of rapid distortion. This regime, first analysed by [Batchelor & Proudman \(1954\)](#) and [Townsend \(1976\)](#) and developed for flow around obstacles by [Hunt \(1973\)](#) and [Goldstein \(1978\)](#), underlies a broad range of engineering and natural flows: the relaminarisation and anisotropy of grid turbulence passing through wind-tunnel contractions ([Mills & Corrsin 1959](#); [Tucker & Reynolds 1968](#); [Lee & Reynolds 1985](#)), the amplification of turbulence across shock waves in compressible aerodynamics ([Ribner 1954](#)), the response of atmospheric turbulence to terrain and to bluff obstacles ([Hunt & Carruthers 1990](#); [Britter *et al.* 1979](#)), and the distortion of oncoming turbulence as it approaches the stagnation region of a lifting surface ([Hunt & Graham 1978](#); [Goldstein & Atassi 1976](#)). In each case the turbulence that emerges from the strained region differs—in intensity, anisotropy and spectral shape—from the turbulence that entered it, and that difference is often the quantity of practical interest.

Capturing this distortion at a cost compatible with parametric study is, however, difficult. Scale-resolving simulation—direct numerical simulation ([Lee & Reynolds 1985](#)) or large-eddy simulation—represents the distortion directly but at a cost that precludes broad sweeps across turbulence intensity, length-scale ratio and Reynolds number. Linear rapid-distortion theory (RDT) provides an analytical description of the mean-strain distortion ([Hunt & Carruthers 1990](#); [Sagaut & Cambon 2008](#)) and is exact in the rapid limit, but it is linear by construction: in its tractable homogeneous form it neglects the nonlinear inter-scale transfer and the slow return-to-isotropy that act over any finite strain duration, so it cannot be relied upon once the accumulated strain is moderate. Second-moment (Reynolds-stress) closures restore the nonlinear relaxation but operate only at the level of single-point moments, discarding the spectral and scale information that the distortion redistributes ([Launder *et al.* 1975](#); [Pope 2000](#)). What is missing, between the prohibitive cost of scale-resolving simulation and the missing physics or missing scales of the analytical and moment-level descriptions, is a reduced-order model that *evolves* strained turbulence with full scale resolution, retaining both the rapid distortion and the nonlinear relaxation, at negligible cost.

A concrete and demanding instance of this general problem—and the application that motivates the present development—is the prediction of turbulence–airfoil interaction noise. When a lifting surface encounters oncoming turbulence, the unsteady upwash induces a fluctuating loading that radiates as broadband sound; for integral scales large compared with the leading-edge radius the source is concentrated at the leading edge, a dominant noise mechanism in axial fans, contra-rotating rotors, wind-turbine and propeller blades, and airframe components in disturbed inflow. The prevailing predictive framework, the analytical theory of [Amiet \(1975\)](#)—which combines the gust-response solution of [Sears \(1941\)](#) with the surface-source formulation of [Curle \(1955\)](#) within the moving-surface analogy of [Ffowcs Williams & Hawkings \(1969\)](#)—obtains the radiated spectrum as a linear functional of the wavenumber–frequency spectrum of the upwash velocity component of the incoming turbulence. The framework has been extended to finite chord,

span and realistic geometry and validated against grid-turbulence experiments (Paterson & Amiet 1977; Roger & Moreau 2010), but in every case *the upwash spectrum that enters the theory is prescribed, not computed*, taken from an idealised isotropic model spectrum—commonly the von Kármán form (von Kármán 1948)—fixed by the upstream intensity and integral scale. Yet the turbulence reaching the leading edge is precisely strained turbulence: as the flow approaches the stagnation region the mean gradient distorts the incident eddies, rendering the upwash spectrum *seen by the edge* anisotropic, scale-dependent and tensorially redistributed relative to its upstream counterpart, while the body additionally imposes a kinematic blocking near the surface (Hunt & Graham 1978). This pre-impact distortion is the same mean-strain distortion described above, occurring in the stagnation region; the standard prediction chain omits it, and because the radiated spectrum is linear in the upwash input, that omission sets a floor on the fidelity of the prediction. A model that supplies the distorted spectrum is therefore of direct value to this application, and it is the distortion problem, not the acoustic stage, that is the subject of this paper.

Among reduced-order representations of turbulence, two further families bear on this gap. Synthetic-turbulence methods—synthetic-eddy and random-particle techniques—efficiently reproduce a *prescribed* spectrum and coherence and are now routine for generating aeroacoustic and inflow boundary conditions (Kim & Haeri 2015), but by construction they impose the spectrum rather than evolving it, and so cannot themselves predict how it is modified under strain. Map-based and stochastic one-dimensional models, by contrast, evolve a resolved field: the linear-eddy model (Kerstein 1991) and the one-dimensional turbulence (ODT) model (Kerstein 1999) carry the full range of scales along a notional line and have proven able to predict spectra and component statistics across a wide range of flows. It is this second family that offers a route to evolving strained turbulence with full scale resolution, and ODT that we develop here.

The one-dimensional turbulence (ODT) model of Kerstein (1999) is, for these reasons, a natural candidate. ODT represents the evolution of a turbulent velocity profile along a single notional line of sight, resolving the full range of length scales in one spatial dimension while modelling turbulent advection as a stochastic sequence of scale-local mapping (“eddy”) events built on the triplet map introduced in the linear-eddy model (Kerstein 1991). Its vector formulation (Kerstein *et al.* 2001) carries all three velocity components and redistributes energy among them through a kernel mechanism that emulates pressure scrambling (Wunsch & Kerstein 2001; Kerstein *et al.* 2001), reproducing inter-component transfer and return to isotropy; the model has been generalised to variable density (Ashurst & Kerstein 2005), to efficient adaptive-mesh and spatially developing implementations (Lignell *et al.* 2013; Stephens & Lignell 2021), to cylindrical and spherical geometries (Lignell *et al.* 2018), and embedded as a subgrid closure in a three-dimensional large-eddy framework (Schmidt *et al.* 2010). Across free-shear and wall-bounded flows it reproduces inertial-range scaling and component statistics at a small fraction of the cost of three-dimensional simulation, and it has been coupled to acoustic analogies to estimate the far-field sound of low-Mach-number turbulent jets (Sharma *et al.* 2022; Medina Méndez *et al.* 2023)—demonstrating that its resolved component spectra and fluctuating Reynolds stresses can be supplied to an external theory. ODT is, in short, an accurate and inexpensive generator of scale-resolved turbulence statistics, of exactly the kind that a model of strained turbulence must deliver.

Standard ODT cannot, however, represent distortion by a mean strain. The model contains no mechanism for the action of an imposed mean velocity gradient on the fluctuating field: its pressure treatment, embodied in the scrambling kernel, models only the slow, turbulence-driven return to isotropy and not the rapid, mean-strain-driven redistribution that any strained flow undergoes. The apparent remedy—importing the linear RDT amplitude

equation, in which the pressure response is the solenoidal projection $P_{ij}(\mathbf{k}) = \delta_{ij} - k_i k_j / k^2$ acting in three-dimensional wavenumber space—is fundamentally incompatible with the one-dimensional model, because the projection requires the full wavevector $\mathbf{k} = (k_1, k_2, k_3)$, whereas a single ODT line resolves only the wavenumber along its own coordinate. Reconciling the inherently three-dimensional and non-local pressure physics of rapid distortion with the one-dimensional and local structure of ODT is the central modelling problem addressed in this paper.

We present a strain-coupled formulation of one-dimensional turbulence that incorporates rapid distortion by an imposed mean velocity gradient while preserving the structural foundations of the model. The formulation rests on a decomposition of the pressure–strain interaction into its rapid and slow parts (Pope 2000): the slow, turbulence-driven part is retained as the existing ODT scrambling kernel, while the rapid, mean-strain-driven part is introduced as a new component-redistribution operator. Rather than evaluating the three-dimensional projection, that operator is constructed so that the component-energy evolution of the line reproduces the exact homogeneous-RDT solution for an imposed mean strain, rendering the formulation provably RDT-consistent in the homogeneous limit while remaining expressed entirely in quantities available on the line. The mean-strain production enters as a continuous forcing of the velocity components, which we show preserves the measure-preservation, continuity and conservation properties of the triplet-map mechanism; the kinematic compression of wavenumbers under strain is captured by a consistent deformation of the ODT domain, connecting the formulation to its adaptive-mesh implementation. The result is a reduced-order, scale-resolved model of strained turbulence that retains the rapid distortion and the nonlinear relaxation together and delivers, as output, the distorted and anisotropic velocity spectrum. We further address the question of principle that the construction raises—whether the rapid pressure–strain, exactly a nonlocal three-dimensional object, can be represented consistently on one line at all—and show that it collapses to a closed, single-line form under one explicit and physically grounded assumption. The leading-edge-noise application that motivates the work is the natural consumer of this output: the distorted upwash spectrum is precisely the input that Amiet’s theory prescribes but does not compute. We develop the distortion model and its validation here and defer the coupling to the acoustic stage—together with the surface blocking and spanwise coherence that a complete leading-edge prediction requires—to a companion paper.

The remainder of the paper is organised as follows. Section 2 reviews the elements of standard ODT required for the present development and the rapid/slow structure of the pressure–strain interaction. Section 3 derives the strain-coupled formulation, establishes the preservation of the model’s structural invariants under mean-strain forcing, and constructs the RDT-consistent rapid operator. Section 4 verifies the formulation against analytical homogeneous-RDT solutions for the canonical strains of plane strain and axisymmetric contraction, examines the emergent distorted spectrum, and validates the model against the strained-turbulence experiment of Chen *et al.* (2006) and the direct numerical simulations of Lee & Reynolds (1985). Section 5 addresses the question of principle, establishing the condition under which the rapid pressure–strain term closes on a single line. Section 6 concludes and outlines the route, through the acoustic coupling and the surface blocking, towards the leading-edge-noise application.

2. Background

This section collects the elements on which the strain-coupled formulation of § 3 is built. Section 2.1 summarises standard one-dimensional turbulence (ODT) in the notation used

throughout; § 2.2 recalls the rapid/slow decomposition of the pressure–strain interaction and the linear rapid-distortion equations that the new formulation is required to reproduce. Nothing in this section is original; readers familiar with Kerstein (1999); Kerstein *et al.* (2001) and with second-moment closure (Pope 2000) may proceed to § 3 after noting the notation and the correspondence stated in § 2.2.

2.1. Standard one-dimensional turbulence

ODT evolves a three-component velocity field $u_i(y, t)$, $i = 1, 2, 3$, and any number of scalar fields $\theta(y, t)$, defined on a one-dimensional domain parameterised by the spatial coordinate y , which is identified with the coordinate direction x_2 (Kerstein *et al.* 2001). The fields represent individual realisations: ensemble statistics are accumulated over many independent simulations rather than carried by the fields themselves. The fields evolve by two interleaved mechanisms—continuous molecular transport and a stochastic sequence of instantaneous *eddy events*.

2.1.1. Molecular transport

Between events the fields obey ordinary one-dimensional diffusion,

$$\left(\partial_t - \nu \partial_y^2\right) u_i(y, t) = 0, \quad \left(\partial_t - \kappa \partial_y^2\right) \theta(y, t) = 0, \quad (2.1)$$

with kinematic viscosity ν and scalar diffusivity κ (Schmidt number $Sc = \nu/\kappa$). In a constant-property, constant-density medium this is the only continuous evolution; the variable-density generalisation is given by Ashurst & Kerstein (2005).

2.1.2. Eddy events: the triplet map and kernel

An eddy event models the effect of a single turbulent overturn. Because a continuum rotational motion cannot be represented on a line, the event is implemented as an instantaneous, measure-preserving rearrangement of the profiles together with an additive velocity change,

$$u_i(y) \longrightarrow u_i(f(y)) + c_i K(y), \quad \theta(y) \longrightarrow \theta(f(y)). \quad (2.2)$$

The map f is the *triplet map* (Kerstein 1991, 1999), defined on an interval $[y_0, y_0 + l]$ of size l by

$$f(y) = y_0 + \begin{cases} 3(y - y_0), & y_0 \leq y \leq y_0 + \frac{1}{3}l, \\ 2l - 3(y - y_0), & y_0 + \frac{1}{3}l \leq y \leq y_0 + \frac{2}{3}l, \\ 3(y - y_0) - 2l, & y_0 + \frac{2}{3}l \leq y \leq y_0 + l, \\ y - y_0, & \text{otherwise,} \end{cases} \quad (2.3)$$

written here through its inverse, so that fluid at $f(y)$ is placed at y . The map compresses the interval to one-third of its length, inserts three copies, and reverses the central copy; it leaves the field outside $[y_0, y_0 + l]$ unchanged. The triplet map satisfies the three structural requirements that underpin the model: it is *measure preserving* (the non-local analogue of a divergence-free rearrangement), *continuous* (it introduces no discontinuities into the fields), and *scale local* (it changes property gradients by at most a factor of order unity) (Kerstein *et al.* 2001). These three properties are exactly the invariants whose preservation under an imposed mean strain is established in § 3.

The additive term in (2.2) is built from the *kernel*

$$K(y) \equiv y - f(y), \quad (2.4)$$

which is non-zero only within the eddy interval and satisfies the two identities used below,

$$\int K(y) dy = 0, \quad \int K^2(y) dy = \frac{4}{27} l^3. \quad (2.5)$$

206 The first guarantees that the additive change conserves the y -integrated momentum of each
 207 velocity component at constant density; the second fixes the energetics of the redistribution.
 208 The amplitudes c_i are determined event-by-event by the pressure-scrumbling rule described
 209 next.

210 2.1.3. Pressure scrambling and component-energy redistribution

The additive kernel term in (2.2) is the model surrogate for the pressure-mediated redistribution of energy among velocity components, an effect absent from the original single-component formulation (Kerstein 1999) and introduced through the kernel by Wunsch & Kerstein (2001); Kerstein *et al.* (2001). Using measure preservation and (2.5), the change in the kinetic energy of component i produced by an event is

$$\Delta E_i = \rho_0 l^2 c_i \left(u_{i,K} + \frac{2}{27} l c_i \right), \quad u_{i,K} \equiv \frac{1}{l^2} \int u_i(f(y)) K(y) dy, \quad (2.6)$$

where ρ_0 is the (constant) density per unit length, which does not affect the constant-density applications considered here. The amplitudes are fixed by requiring that the event redistribute energy among components while conserving the total, $\sum_i \Delta E_i = 0$, and that the rule be invariant under permutation of the component indices. These requirements give

$$\Delta E_i = \alpha \sum_j T_{ij} Q_j, \quad \mathbf{T} = \frac{1}{2} \begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix}, \quad Q_i \equiv \frac{27}{8} \rho_0 l u_{i,K}^2, \quad (2.7)$$

in which Q_i is the *available kinetic energy* of component i within the eddy and $\alpha \in [0, 1]$ is the model parameter setting the fraction of extractable energy transferred towards isotropy (Kerstein *et al.* 2001). Combining (2.6) and (2.7) determines the amplitudes,

$$c_i = \frac{27}{4l} \left[-u_{i,K} + \text{sgn}(u_{i,K}) \sqrt{u_{i,K}^2 + \alpha \sum_j T_{ij} u_{j,K}^2} \right], \quad (2.8)$$

211 which is real for $0 \leq \alpha \leq 1$ and vanishes as $\alpha \rightarrow 0$. The transfer matrix $\mathbf{T}$ acts to equalise
 212 the component energies; the construction (2.7)–(2.8) is therefore a model of the *return-*
 213 *to-isotropy* (slow) part of the pressure–strain interaction, a point made precise in § 2.2.
 214 Standard ODT contains no counterpart of the rapid, mean-strain-driven part.

215 2.1.4. Eddy selection

The event sequence is a Poisson process whose rate reflects the instantaneous state of the flow. Each candidate eddy (y_0, l) is assigned a time scale $\tau(y_0, l; t)$ built from the available energy of the velocity component aligned with the domain,

$$\left(\frac{l}{\tau} \right)^2 \sim u_{2,K}^2 + \alpha \sum_j T_{2j} u_{j,K}^2 - Z \frac{v^2}{l^2}, \quad (2.9)$$

the final term being a viscous penalty, controlled by the order-unity parameter Z , that suppresses eddies slower than their viscous time. The corresponding event-rate distribution

is

$$\lambda(y_0, l; t) = \frac{C}{l^2 \tau(y_0, l; t)}, \quad (2.10)$$

with rate constant C ; eddies for which the right-hand side of (2.9) is negative are forbidden. Crucially, the inertial-range cascade and the $E(k) \sim k^{-5/3}$ scaling are *outcomes* of repeated sampling from (2.10), not inputs to it (Kerstein 1999). The three constants C , Z and a large-eddy suppression parameter are the only tunable inputs of the model (Stephens & Lignell 2021).

2.1.5. Statistical observables

ODT yields single- and multi-point moments, probability density functions, conditional statistics, power spectra and Lagrangian statistics (Kerstein 1999). For the present work the relevant observable is the single-component velocity spectrum—in particular the spectrum and variance of the component normal to the airfoil, which are second moments of the velocity field and are obtained directly from the realisations. A distinction emphasised by Kerstein *et al.* (2001) must, however, be respected: because the line velocity does not itself advect fluid, an off-diagonal Reynolds stress such as $\langle u'_1 u'_2 \rangle$ *cannot* be interpreted as a momentum flux when formed as a product of the line velocity components; such advective fluxes are instead evaluated by monitoring the transport induced by the eddy events. This distinction constrains how the anisotropy produced by the formulation of § 3 is to be diagnosed and validated (§ 4): diagonal component spectra are read from the fields, whereas shear stresses are read from eddy-induced fluxes.

2.2. Rapid and slow pressure–strain, and linear rapid distortion

The Reynolds-stress transport equation for a fluctuating field u'_i about a mean U_i with constant velocity gradient $A_{ij} \equiv \partial U_i / \partial x_j$ reads, in homogeneous turbulence,

$$\frac{d\langle u'_i u'_j \rangle}{dt} = \mathcal{P}_{ij} + \Pi_{ij} - \varepsilon_{ij}, \quad (2.11)$$

with production, pressure–strain and dissipation

$$\mathcal{P}_{ij} = -\langle u'_i u'_k \rangle A_{jk} - \langle u'_j u'_k \rangle A_{ik}, \quad \Pi_{ij} = \left\langle \frac{p'}{\rho} (\partial_j u'_i + \partial_i u'_j) \right\rangle, \quad \varepsilon_{ij} = 2\nu \langle \partial_k u'_i \partial_k u'_j \rangle. \quad (2.12)$$

Taking the divergence of the fluctuating momentum equation gives the pressure as the solution of a Poisson equation whose source splits into a part linear in the mean velocity gradient and a part quadratic in the fluctuations,

$$\frac{1}{\rho} \nabla^2 p' = \underbrace{-2 A_{lm} \partial_l u'_m}_{\text{rapid}} - \underbrace{\partial_l \partial_m (u'_l u'_m - \langle u'_l u'_m \rangle)}_{\text{slow}}. \quad (2.13)$$

The pressure–strain correlation inherits this decomposition, $\Pi_{ij} = \Pi_{ij}^{(r)} + \Pi_{ij}^{(s)}$, into a *rapid* part driven by the mean strain and a *slow* part driven by the turbulence alone (Pope 2000). The slow part relaxes the Reynolds stress towards isotropy; the simplest closure is that of Rotta, $\Pi_{ij}^{(s)} = -C_1(\varepsilon/k)(\langle u'_i u'_j \rangle - \frac{2}{3}k \delta_{ij})$, with $k = \frac{1}{2}\langle u'_i u'_i \rangle$. It is this return-to-isotropy mechanism that the ODT kernel (2.7)–(2.8) reproduces in model form. The rapid part is linear in A_{lm} and is determined, in homogeneous turbulence, by the velocity-spectrum tensor $\Phi_{ij}(\mathbf{k})$; the corresponding “isotropisation of production” closure

242 is $\Pi_{ij}^{(r)} = -C_2(\mathcal{P}_{ij} - \frac{2}{3}\mathcal{P}\delta_{ij})$, with $\mathcal{P} = \frac{1}{2}\mathcal{P}_{ii}$ (Pope 2000). Standard ODT represents
 243 neither $\mathcal{P}_{ij}$ nor $\Pi_{ij}^{(r)}$.

For a rapidly applied strain—formally, when the total strain accumulates in a time short compared with the eddy-turnover time—the rapid part can be computed exactly from linear theory (Batchelor & Proudman 1954; Hunt 1973; Hunt & Carruthers 1990). Writing the fluctuation as a sum of Fourier modes that follow the mean flow, each mode of (time-dependent) wavevector $\mathbf{k}(t)$ and amplitude $\hat{u}_i(\mathbf{k}, t)$ evolves, in the inviscid rapid limit, according to

$$\frac{d\mathbf{k}_i}{dt} = -A_{ji}k_j, \quad \frac{d\hat{u}_i}{dt} = \left(\frac{2k_ik_m}{k^2} - \delta_{im} \right) A_{mn}\hat{u}_n, \quad (2.14)$$

where $k^2 = k_mk_m$. The bracketed operator combines the production $-A_{im}\hat{u}_m$ with the rapid pressure response, the latter being the solenoidal projection $P_{ij}(\mathbf{k}) = \delta_{ij} - k_ik_j/k^2$ applied to the strained field. The factor of two in (2.14) is fixed by the requirement that the motion remain divergence free: using both relations in (2.14),

$$\frac{d}{dt}(k_i\hat{u}_i) = \dot{k}_i\hat{u}_i + k_i\dot{\hat{u}}_i = -A_{ji}k_j\hat{u}_i + \left(\frac{2k^2k_m}{k^2} - k_m \right) A_{mn}\hat{u}_n = 0, \quad (2.15)$$

244 so that $k_i\hat{u}_i = 0$ is preserved; any other coefficient violates continuity. Equations (2.14)
 245 integrate the spectral tensor through the distortion and thereby determine the rapid
 246 pressure–strain $\Pi_{ij}^{(r)}$ and the component-energy evolution exactly.

247 Two features of (2.14) are central to what follows. First, the pressure response is the
 248 operator $P_{ij}(\mathbf{k})$, which is *non-local* (it couples the components through the full wavevector)
 249 and intrinsically *three-dimensional* (it requires all of k_1, k_2, k_3). A single ODT line resolves
 250 only the wavenumber along its own coordinate, so (2.14) cannot be evaluated on the line as
 251 written; reconciling this incompatibility is the subject of § 3. Second, the decomposition
 252 $\Pi_{ij} = \Pi_{ij}^{(r)} + \Pi_{ij}^{(s)}$ furnishes the organising principle of the new formulation: the slow
 253 part is already carried by the ODT kernel (2.7)–(2.8), and it remains only to supply the
 254 production $\mathcal{P}_{ij}$ and a rapid redistribution consistent with (2.14), expressed in quantities
 255 available on the line.

256 3. Strain-coupled one-dimensional turbulence

257 The linear rapid-distortion equations (2.14) contain two distinct ingredients: an evolution
 258 of the Fourier *amplitudes* $\hat{u}_i(\mathbf{k}, t)$, comprising production and the solenoidal pressure
 259 projection, and an evolution of the *wavevectors* $\mathbf{k}(t)$ that carries the kinematic straining
 260 of scales. The amplitude equation is the obstruction identified in § 2.2: its pressure term
 261 requires the full three-dimensional wavevector and is therefore not directly representable
 262 on a one-dimensional line. Our strategy is to split the distortion physics according to this
 263 structure and to assign each part to the ODT mechanism that can carry it faithfully. The
 264 production, which is local in physical space, is imposed as a continuous forcing of the
 265 line velocity (§ 3.1); the rapid pressure–strain, which is not, is supplied as a continuous,
 266 energy-conserving redistribution operator constructed to reproduce the homogeneous
 267 rapid-distortion evolution of the component energies under an explicit spectral closure
 268 (§ 3.2); and the wavevector kinematics are recovered by a consistent straining of the
 269 ODT domain (§ 3.3). The slow, turbulence-driven pressure–strain and the inertial cascade
 270 remain the discrete eddy events of standard ODT, unchanged. Section 3.4 assembles the
 271 formulation and identifies the parameter governing the balance between the rapid and slow
 272 mechanisms.

Throughout, $A_{ij} = \partial U_i / \partial x_j$ denotes the imposed mean velocity gradient, taken uniform across the line at each instant and varying along the mean trajectory; the medium is incompressible, $A_{ii} = 0$. The line carries the fluctuation field $u_i(y, t)$ about this mean. We write $R_{ij} \equiv \overline{u_i u_j}$ for the Reynolds-stress tensor, where the overline denotes the average over the statistically homogeneous directions (equivalently, the ensemble of independent realisations), and $k_t \equiv \frac{1}{2} R_{ii}$ for the turbulent kinetic energy. In the homogeneous configurations used to establish and test the formulation (§4) the spatial line average and the ensemble average coincide.

3.1. The strain-coupled line equation and its structural admissibility

In rapid-distortion theory the production term of the fluctuation momentum equation, $-A_{ij}u_j$, is a pointwise linear combination of the local velocity components and is therefore exactly representable on the line. Between eddy events we replace the molecular evolution (2.1) by

$$\partial_t u_i = \nu \partial_y^2 u_i - A_{ij} u_j + B_{ij} u_j, \quad (3.1)$$

in which $-A_{ij}u_j$ is the production and $B_{ij}u_j$ is the rapid pressure-strain operator constructed in §3.2; we defer B_{ij} and treat the production term first. The eddy events (2.2)–(2.8) are retained without modification. The evolution is thus advanced by operator splitting between a continuous phase governed by (3.1) and a discrete sequence of events sampled from (2.10).

A formulation is admissible only if the additional forcing does not violate the structural properties on which ODT rests. We establish this directly.

PROPOSITION 1 (STRUCTURAL ADMISSIBILITY). *The strain forcing in (3.1) preserves the four structural properties of standard ODT: (i) the measure-preservation, (ii) continuity and (iii) scale-locality of the triplet map, and (iv) the kernel identities (2.5).*

Proof. Properties (i)–(iv) are properties of the discrete event, that is, of the map f (2.3) and the kernel K (2.4). Because the advancement is split into a continuous phase (3.1) and the unaltered discrete event, the map and kernel are applied exactly as in standard ODT; properties (i), (iii) and (iv), which are geometric statements about f and K on a fixed domain, therefore hold verbatim. It remains to verify that the continuous phase does not destroy the continuity of the fields, which underlies (ii). For a mean gradient uniform along the line, the linear, y -independent operator $\mathcal{A}_{ij} \equiv -A_{ij} + B_{ij}$ acts pointwise, and the solution of the advective part of (3.1) over a sub-step $[t_0, t]$ is

$$u_i(y, t) = \left[\exp \int_{t_0}^t \mathcal{A} dt' \right]_{ij} u_j(y, t_0), \quad (3.2)$$

a matrix that is independent of y . It maps continuous (respectively, C^∞) profiles to continuous (C^∞) profiles and introduces no spatial discontinuity, while the molecular term $\nu \partial_y^2 u_i$ is smoothing. Continuity is preserved, establishing (ii). The forcing displaces no fluid along y , so it leaves the Lebesgue measure of the domain unchanged; the deliberate straining of the domain is treated separately in §3.3. $\square$

The energetic role of the forcing follows from (3.1). At fixed domain length, the production term evolves the Reynolds stress as

$$\left. \frac{dR_{ij}}{dt} \right|_{\text{prod}} = \overline{(\dot{u}_i u_j + u_i \dot{u}_j)} = -A_{ik} R_{kj} - A_{jk} R_{ki} \equiv \mathcal{P}_{ij}, \quad (3.3)$$

which is *exactly* the Reynolds-stress production tensor of (2.12); the corresponding kinetic-energy production is $\mathcal{P} \equiv \frac{1}{2} \mathcal{P}_{ii} = -A_{ij} R_{ij}$. The production term thus carries no model

assumption. With diffusion supplying dissipation through (2.1), the kinetic-energy budget of the formulation separates cleanly into an exact production by the mean strain, a redistribution that conserves energy (the kernels), and a dissipation (molecular diffusion). This is the structure of the exact Reynolds-stress equation (2.11), and it is the reason the forcing is admissible: it adds the one term—production—that standard ODT lacks, without disturbing the terms it already represents.

3.2. The RDT-consistent rapid redistribution operator

3.2.1. Why production alone is insufficient

Were the continuous evolution to consist of production and diffusion only ($B_{ij} = 0$), the line would reproduce $dR_{ij}/dt = \mathcal{P}_{ij}$, whereas the exact inviscid rapid-distortion balance is $dR_{ij}/dt = \mathcal{P}_{ij} + \Pi_{ij}^{(r)}$. The omission is not small. For turbulence that is isotropic at the onset of straining, the classical result derived below gives $\Pi_{ij}^{(r)} = -\frac{3}{5}\mathcal{P}_{ij}$, so that the true initial distortion rate is $\frac{2}{5}\mathcal{P}_{ij}$: production acting alone overstates the rate at which anisotropy develops by a factor of $\frac{5}{2}$. For sustained strong strain, neglect of the pressure response also permits a component energy to grow without the bound that incompressibility imposes, violating realisability. Representing $\Pi_{ij}^{(r)}$ is therefore essential, not optional.

3.2.2. The exact rapid pressure–strain and the closure problem

The exact rapid pressure–strain follows from the amplitude equation (2.14). Writing the operator there as $d\hat{u}_i/dt = M_{in}\hat{u}_n$ with $M_{in} = -A_{in} + 2(k_ik_m/k^2)A_{mn}$, and noting that the incompressible wavevector dynamics $\dot{k}_i = -A_{ji}k_j$ preserve the measure $d\mathbf{k}$ (its phase-space divergence is $-A_{ii} = 0$), the spectral tensor $\Phi_{ij} = \langle \hat{u}_i^* \hat{u}_j \rangle$ yields

$$\frac{dR_{ij}}{dt} = \int (M_{in}\Phi_{nj} + M_{jn}\Phi_{in}) d\mathbf{k} = \mathcal{P}_{ij} + \Pi_{ij}^{(r)}, \quad (3.4)$$

in which the production (3.3) has been separated from the pressure contribution,

$$\Pi_{ij}^{(r)} = 2A_{mn}(\mathcal{R}_{ijmn} + \mathcal{R}_{jimn}), \quad \mathcal{R}_{ijmn} \equiv \int \frac{k_ik_mk_n}{k^2} \Phi_{nj}(\mathbf{k}) d\mathbf{k}. \quad (3.5)$$

The tensor $\Pi_{ij}^{(r)}$ is traceless: using the solenoidality $k_n\Phi_{nj} = 0$ (a consequence of $k_i\hat{u}_i = 0$),

$$\Pi_{ii}^{(r)} = 4A_{mn} \int \frac{k_ik_mk_n}{k^2} \Phi_{ni} d\mathbf{k} = 4A_{mn} \int \frac{k_m}{k^2} (k_i\Phi_{ni}) d\mathbf{k} = 0, \quad (3.6)$$

so the rapid pressure–strain redistributes energy among components without changing the total, in keeping with the energy-conserving character of the ODT kernel mechanism.

Equation (3.5) exposes the central difficulty. The rapid pressure–strain is a functional of the spectral tensor $\Phi_{nj}(\mathbf{k})$ through the orientation-weighted integral $\mathcal{R}_{ijmn}$; it is *not* expressible in terms of the single-point Reynolds stress R_{ij} alone. This is the well-known closure problem of the rapid pressure–strain, common to every single-point model (Crow 1968; Launder *et al.* 1975; Pope 2000); it is intrinsic to the physics and not a deficiency peculiar to the present approach. A one-dimensional line carries R_{ij} and the one-dimensional spectrum $\Phi_{ij}(k_2)$ along its own coordinate, but it does not carry the distribution of energy over wavevector *orientation* that $\mathcal{R}_{ijmn}$ requires. A spectral-tensor closure is therefore unavoidable, exactly as in second-moment closure; we make it explicit.

327 3.2.3. *Baseline isotropic closure*

The minimal closure assumes the spectral tensor to be instantaneously isotropic, $\Phi_{nj}(\mathbf{k}) = \frac{E(k)}{4\pi k^2}(\delta_{nj} - k_n k_j / k^2)$ with $\int E(k) dk = k_t$. Writing $e_i \equiv k_i / k$ and using the angular averages over the unit sphere $\overline{e_i e_m} = \frac{1}{3} \delta_{im}$ and $\overline{e_i e_m e_n e_j} = \frac{1}{15}(\delta_{im} \delta_{nj} + \delta_{in} \delta_{mj} + \delta_{ij} \delta_{mn})$,

$$\mathcal{R}_{ijmn}^{\text{iso}} = k_t \left[\frac{4}{15} \delta_{im} \delta_{nj} - \frac{1}{15} (\delta_{in} \delta_{mj} + \delta_{ij} \delta_{mn}) \right]. \quad (3.7)$$

Substituting (3.7) into (3.5) and using $A_{nn} = 0$ gives, with $S_{ij} \equiv \frac{1}{2}(A_{ij} + A_{ji})$,

$$\Pi_{ij}^{(r),\text{iso}} = 2k_t \left[\frac{4}{15} A_{ij} - \frac{1}{15} A_{ji} \right] + 2k_t \left[\frac{4}{15} A_{ji} - \frac{1}{15} A_{ij} \right] = \frac{4}{5} k_t S_{ij}. \quad (3.8)$$

For isotropic $R_{ij} = \frac{2}{3} k_t \delta_{ij}$ the production (3.3) is $\mathcal{P}_{ij} = -\frac{4}{3} k_t S_{ij}$, so that (3.8) is equivalent to

$$\Pi_{ij}^{(r),\text{iso}} = -\frac{3}{5} \mathcal{P}_{ij}, \quad (3.9)$$

328 the exact response of initially isotropic turbulence to a weak rapid strain (Crow 1968).
 329 Equation (3.9) is the content of the isotropisation-of-production model with the constant
 330 $C_2 = \frac{3}{5}$ (Pope 2000); here it is obtained, with no adjustable constant, as the isotropic limit
 331 of the rapid-distortion integral (3.5).

332 When the turbulence is anisotropic, (3.7) no longer holds and $\mathcal{R}_{ijmn}$ must be modelled
 333 as a tensor function of R_{ij} ; the general representation linear in the anisotropy is that of
 334 Launder *et al.* (1975), which introduces calibrated constants and to which the present
 335 construction reduces in form. We adopt (3.7) as the baseline and treat the anisotropic
 336 correction as a modelling option, returning to its assessment in § 4.

337 3.2.4. *Realisation on the line*

Given the closed rapid pressure–strain $\Pi_{ij}^{(r)}$ —symmetric and traceless by (3.6)—we require a pointwise operator B_{ij} on the line whose action reproduces it. Demanding that the operator evolve the Reynolds stress by exactly $\Pi_{ij}^{(r)}$,

$$B_{ik} R_{kj} + B_{jk} R_{ki} = (\mathbf{B}\mathbf{R} + \mathbf{R}\mathbf{B})_{ij} = \Pi_{ij}^{(r)}, \quad (3.10)$$

where the second equality holds for symmetric $\mathbf{B}$. Equation (3.10) is a continuous Lyapunov equation; for a positive-definite Reynolds stress $\mathbf{R} \succ 0$ it possesses a unique symmetric solution $\mathbf{B}$. The solution conserves kinetic energy automatically: contracting (3.10) gives $2B_{ij} R_{ij} = \Pi_{ii}^{(r)} = 0$, so that the rate of working of the operator on the line, $\overline{B_{ij} u_i u_j} = B_{ij} R_{ij}$, vanishes. In the isotropic case $\mathbf{R} = \frac{2}{3} k_t \mathbf{I}$, (3.10) together with (3.8) reduces to $\frac{4}{3} k_t \mathbf{B} = \frac{4}{5} k_t \mathbf{S}$, that is, the closed form

$$B_{ij} = \frac{3}{5} S_{ij}, \quad \mathcal{A}_{ij} = -A_{ij} + \frac{3}{5} S_{ij}. \quad (3.11)$$

338 The operator B_{ij} is evaluated from the running Reynolds stress; it is applied continuously,
 339 paced by the mean strain, exactly as is the production. This is deliberate: the rapid pressure–
 340 strain is, by construction, the part of the redistribution driven by the mean velocity
 341 gradient and therefore acts at the strain rate, in contrast to the slow, turbulence-driven
 342 redistribution that the eddy kernel (2.7)–(2.8) supplies at the eddy rate. The two mechanisms
 343 are thereby cleanly separated: strain-driven effects (production and rapid pressure–strain)
 344 are continuous and enter through the operator $\mathcal{A}_{ij}$, while turbulence-driven effects (the
 345 inertial cascade and the slow return to isotropy) remain the discrete events. The uniform,
 346 scale-independent action of B_{ij} is the faithful single-point reduction of the rapid pressure

response, which in (2.14) acts on every Fourier mode through the projection $P_{ij}(\mathbf{k})$; the distribution of energy across scales continues to be governed by diffusion, the triplet-map cascade and the domain straining of § 3.3.

We can now state precisely the sense in which the formulation is rapid-distortion consistent.

PROPOSITION 2 (RAPID-DISTORTION CONSISTENCY). *Let B_{ij} be defined by the Lyapunov equation (3.10) with $\Pi_{ij}^{(r)}$ evaluated from a prescribed spectral-tensor closure. Then the continuous evolution (3.1) reproduces the homogeneous rapid-distortion evolution of the Reynolds stress (3.4) exactly for that closure. With the isotropic closure (3.7), the reproduction is exact at the onset of distortion of initially isotropic turbulence and contains no adjustable constant, recovering the classical result (3.9).*

Proof. By (3.3) the production term of (3.1) contributes $\mathcal{P}_{ij}$ to dR_{ij}/dt , and by construction (3.10) the operator B_{ij} contributes $\Pi_{ij}^{(r)}$. Their sum is the right-hand side of (3.4). For initially isotropic turbulence $\Pi_{ij}^{(r)}$ equals the exact value (3.9) by (3.7)–(3.8). $\square$

The scope of Proposition 2 must be stated as carefully as the result. The consistency is established at the level of the *component energies* R_{ij} and is *conditional on the spectral closure*: the isotropic closure is exact only at the onset of distortion, and its accuracy for strongly anisotropic states is that of the isotropisation-of-production approximation it reproduces. The *spectral* distribution of the distorted energy is not enforced by (3.1); it emerges from the interaction of the strain operator with diffusion, the cascade and the domain straining, and is therefore a genuine prediction of the model, to be tested rather than assumed (§ 4). We make no claim that the formulation reproduces the full spectral evolution of rapid-distortion theory, which no single-point model can; the claim is the weaker, provable one of Proposition 2.

3.3. Wavevector kinematics by domain straining

The second ingredient of (2.14), the wavevector evolution $dk_i/dt = -A_{ji}k_j$, carries the kinematic compression and stretching of scales by the mean strain and is independent of the amplitude evolution; in the method-of-characteristics reading of rapid-distortion theory, the wavevectors are the characteristics along which the amplitudes evolve. On the line, aligned with the coordinate x_2 , the relevant wavenumber is k_2 . We restrict attention to the case in which the line is aligned with a principal axis of the mean strain, so that $A_{12} = A_{32} = 0$ and a material element of the line remains a coordinate line; this holds, in particular, for the irrotational mean flow of a stagnation region, for which A_{ij} is symmetric and its eigenvectors define such axes. Then

$$\frac{dk_2}{dt} = -A_{22} k_2. \quad (3.12)$$

A material element of the line of length ℓ stretches with the mean flow as $\dot{\ell}/\ell = A_{22}$; since wavenumbers scale as $k_2 \propto 1/\ell$, a uniform dilatation of the line reproduces (3.12) exactly,

$$\frac{1}{L} \frac{dL}{dt} = A_{22}, \quad \implies \quad k_2(t) = k_2(t_0) \exp\left(-\int_{t_0}^t A_{22} dt'\right), \quad (3.13)$$

where L is the domain length. The dilatation (3.13) is implemented directly within the Lagrangian adaptive-mesh advancement of ODT (Lignell *et al.* 2013), in which cell sizes already evolve, by imposing the additional mean-flow stretching on the mesh.

This step is not redundant with the amplitude operator of §§ 3.1–3.2. Because the Reynolds stress R_{ij} is an intensive (per-unit-length) second moment, it is invariant under the uniform dilatation (3.13): the dilatation relabels the scales that carry the energy without changing the component energies, while the operator $\mathcal{A}_{ij}$ changes the component energies without relabelling scales. The two operations are the physical-space counterparts of the characteristic motion $\mathbf{k}(t)$ and the amplitude evolution $\hat{u}(\mathbf{k}(t), t)$ in (2.14), and together they account for the distinct contributions of mean straining without double counting.

Two limitations are inherent to representing a three-dimensional kinematics on one axis. The transverse wavenumbers k_1 and k_3 also strain under $dk_i/dt = -A_{ji}k_j$, but are not represented on the line; their effect on the redistribution is subsumed in the spectral closure of § 3.2, which already supplies the orientation information the line lacks. A mean flow with a rotational part, or a line not aligned with a principal axis, tilts the material line out of the coordinate direction and couples it to the unresolved directions; this lies outside the present formulation and is admissible only to the extent that the stagnation-region mean flow is irrotational.

3.4. Summary of the formulation

The strain-coupled model advances the line field $u_i(y, t)$ over a sub-step by composing three operations:

- (a) *continuous strain–diffusion phase* on a dilating domain: integrate $\partial_t u_i = \nu \partial_y^2 u_i + \mathcal{A}_{ij} u_j$ with $\mathcal{A}_{ij} = -A_{ij} + B_{ij}$, the production term exact and the rapid term B_{ij} given by (3.10) from the current Reynolds stress and the spectral closure (the baseline being $B_{ij} = \frac{3}{5} S_{ij}$, (3.11)); simultaneously dilate the domain by (3.13);
- (b) *discrete eddy events* sampled from the rate distribution (2.10): the triplet map (2.3) and the slow kernel (2.7)–(2.8), unchanged from standard ODT.

Step (a) carries the mean-strain physics—production, rapid pressure–strain and the kinematic straining of scales—and step (b) carries the turbulence-driven physics—the inertial cascade and the slow return to isotropy. Standard ODT is recovered exactly when $A_{ij} = 0$.

The relative importance of the two steps is governed by the ratio of the mean-strain rate $S \equiv (2S_{ij}S_{ij})^{1/2}$ to the turbulence relaxation rate ε/k_t ,

$$\chi \equiv \frac{S k_t}{\varepsilon}. \quad (3.14)$$

For $\chi \gg 1$ the distortion accumulates faster than the turbulence can respond, the eddy events are negligible over the distortion time, and the formulation reduces to the rapid-distortion limit of step (a), exact under the closure by Proposition 2. For $\chi \ll 1$ the turbulence relaxes between increments of strain, the slow kernel dominates the redistribution, and the model returns to a near-equilibrium anisotropy. The stagnation region of a leading edge is the intermediate regime, $\chi = O(1)$, in which neither limit is uniformly valid and both mechanisms contribute; it is precisely there that a formulation carrying the two together, rather than either alone, is required. The operating point of the simulations reported in § 4.2 sits in this regime, at $\chi = S k_t / \varepsilon \approx 0.8$ (interquartile range 0.7–0.9 over the strained trajectory), confirming that the emergent-spectrum results are obtained where the distinction between the rapid and near-equilibrium limits is physically meaningful. The quantitative behaviour across this range, and the fidelity of the emergent distorted spectrum, are assessed against rapid-distortion solutions and scale-resolving data in § 4.

4. Verification and validation

We separate two questions. *Verification* asks whether the formulation reproduces a known exact solution in the regime where one exists; the only such regime is the rapid-distortion limit, against which § 4.1 tests the formulation analytically. *Validation* asks whether the formulation reproduces the behaviour of the true flow outside that limit; § 4.2 examines the emergent distorted spectrum, and § 4.3–§ 4.4 compare against external reference data—the strained-turbulence experiment of Chen *et al.* (2006) and the direct numerical simulations of Lee & Reynolds (1985)—together with the metrics and the diagnostic value of each possible failure. The analytical benchmarks of § 4.1 are exact and are stated in full; the computed comparisons are reported once the corresponding simulations are completed.

4.1. Verification against rapid-distortion theory

4.1.1. Canonical strains

Two irrotational homogeneous strains both admit exact rapid-distortion solutions and isolate the physics of interest. *Plane strain*,

$$\mathbf{A} = \text{diag}(a, -a, 0), \quad a > 0, \quad (4.1)$$

is the strain of a two-dimensional stagnation region: with x_1 tangential to the surface, x_2 normal to it and x_3 spanwise, the mean flow accelerates away from the stagnation line along x_1 and decelerates towards the surface along x_2 . The component compressed by the strain is the wall-normal (upwash) component u_2 , whose amplification is the pre-impact distortion relevant to leading-edge noise. *Axisymmetric strain*,

$$\mathbf{A} = \text{diag}(-\tfrac{1}{2}\gamma, -\tfrac{1}{2}\gamma, \gamma), \quad \gamma > 0, \quad (4.2)$$

is the strain of a contraction or a three-dimensional stagnation flow and provides an independent test in which two components are amplified and one suppressed. For both, $\mathbf{A}$ is symmetric, so $\mathbf{A} = \mathbf{S}$ and the principal-axis alignment assumed in § 3.3 holds; the ODT line is taken along x_2 .

4.1.2. Onset behaviour: an exact, constant-free check

For turbulence isotropic at the onset of straining, $R_{ij} = \frac{2}{3}k_t\delta_{ij}$, the continuous evolution (3.1) gives, by Proposition 2 and equations (3.3) and (3.9),

$$\left. \frac{dR_{ij}}{dt} \right|_0 = \mathcal{P}_{ij} + \Pi_{ij}^{(r),\text{iso}} = \frac{2}{5}\mathcal{P}_{ij} = -\frac{8}{15}k_t S_{ij}, \quad (4.3)$$

which is the exact rapid-distortion rate for initially isotropic turbulence and contains no adjustable constant. Equation (4.3) is the sharpest verification available: it is a closed-form prediction that the model must satisfy identically, not approximately. Evaluating it for the two canonical strains gives the component rates in table 1. The wall-normal (compressed) component is amplified in both cases; for plane strain the model predicts the upwash growth rate

$$\left. \frac{dR_{22}}{dt} \right|_0 = +\frac{8}{15}k_t a, \quad (4.4)$$

to be compared with the value $\frac{4}{3}k_t a$ that production alone would give: neglect of the rapid pressure–strain overstates the upwash amplification rate by the factor $\frac{5}{2}$ anticipated in § 3.2. A model that passes (4.3) for both strains has its production and rapid pressure–strain terms verified jointly; failure would indicate an error in the operator $\mathcal{A}_{ij}$ or its implementation, isolated from any closure or cascade effect.

	$dR_{11}/dt _0$	$dR_{22}/dt _0$	$dR_{33}/dt _0$
Plane strain (4.1)	$-\frac{8}{15}k_t a$	$+\frac{8}{15}k_t a$	0
Axisymmetric (4.2)	$+\frac{4}{15}k_t \gamma$	$+\frac{4}{15}k_t \gamma$	$-\frac{8}{15}k_t \gamma$

Table 1. Exact onset rates of the component energies from (4.3) for initially isotropic turbulence. The component compressed by the strain (u_2 for plane strain; u_1, u_2 for axisymmetric contraction) is amplified; the stretched component is suppressed.

4.1.3. Finite total strain

Beyond onset the turbulence becomes anisotropic and the rapid pressure–strain (3.5) ceases to be a function of the Reynolds stress alone, so the verification becomes a test of the spectral closure rather than of the construction. We integrate the model moment equation $dR_{ij}/dt = \mathcal{P}_{ij} + \Pi_{ij}^{(r)}$ along each canonical strain with two closures for $\Pi_{ij}^{(r)}$: the baseline isotropic closure (3.7), equivalent to the isotropisation-of-production model with $C_2 = 3/5$ (hereafter IP), and the quasi-isotropic model of Launder *et al.* (1975) (hereafter LRR),

$$\Pi_{ij}^{(r)} = C_2 k_t S_{ij} + C_3 k_t (b_{ik} S_{jk} + b_{jk} S_{ik} - \frac{2}{3} b_{mn} S_{mn} \delta_{ij}) + C_4 k_t (b_{ik} W_{jk} + b_{jk} W_{ik}), \quad (4.5)$$

with $b_{ij} = R_{ij}/2k_t - \frac{1}{3}\delta_{ij}$, mean rotation $W_{ij} = \frac{1}{2}(A_{ij} - A_{ji})$, and constants $C_2 = 4/5$, $C_3 = 7/4$, $C_4 = 131/100$. The coefficient $C_2 = 4/5$ is fixed by the requirement that (4.5) reduce at isotropy ($b_{ij} = 0$) to the exact value $\frac{4}{5}k_t S_{ij}$ of (3.8), so that LRR shares the constant-free onset slope (4.3); the strains considered here are irrotational ($W_{ij} = 0$). The benchmark is the exact rapid-distortion solution, obtained not from a tabulated closed form but by integrating the spectral equations (2.14) over an ensemble of initial wavevector directions on the unit sphere (Gauss–Legendre in $\cos \theta$ by uniform azimuth); for initially isotropic turbulence the modal dynamics depend only on direction, so the Reynolds stress is the spherical average of the evolved modal covariance. This integration is independent of the moment models and reproduces the analytic onset slope (4.3) to within 1.5×10^{-3} (the quadrature error), which serves as a check on the rapid-distortion machinery itself.

Figure 1 reports the resulting component-energy fractions against total strain $e = \int S dt$ for the two strains. Three observations follow. First, at onset all three curves for each component are tangent: the constant-free identity (4.3) is satisfied, and through $e \lesssim 1$ the closures are nearly indistinguishable from the exact solution. This is the numerical counterpart of Proposition 2. Second, as anisotropy accumulates the closures depart from the exact solution, and the LRR closure is uniformly closer than IP: for axisymmetric strain at $e = 4$ the lateral fraction is 0.498 (exact), 0.468 (LRR) and 0.450 (IP), and the axial fraction 0.003, 0.064 and 0.100 respectively. This gap is the spectral-closure error in isolation—the eddy events are inactive in this rapid limit—and it bounds the accuracy attainable from a single rapid operator. Third, and most important to state plainly, for plane strain both closures miss a *qualitative* feature: the exact solution has the spanwise component $\overline{u_3^2}$, which is neither stretched nor compressed, grow monotonically and overtake the upwash component near $e \approx 3.3$, becoming the largest component by $e = 4$ ($\overline{u_3^2}/2k_t = 0.51$ against $\overline{u_2^2}/2k_t = 0.47$), whereas both closures predict $\overline{u_3^2}$ to decay. This redistribution

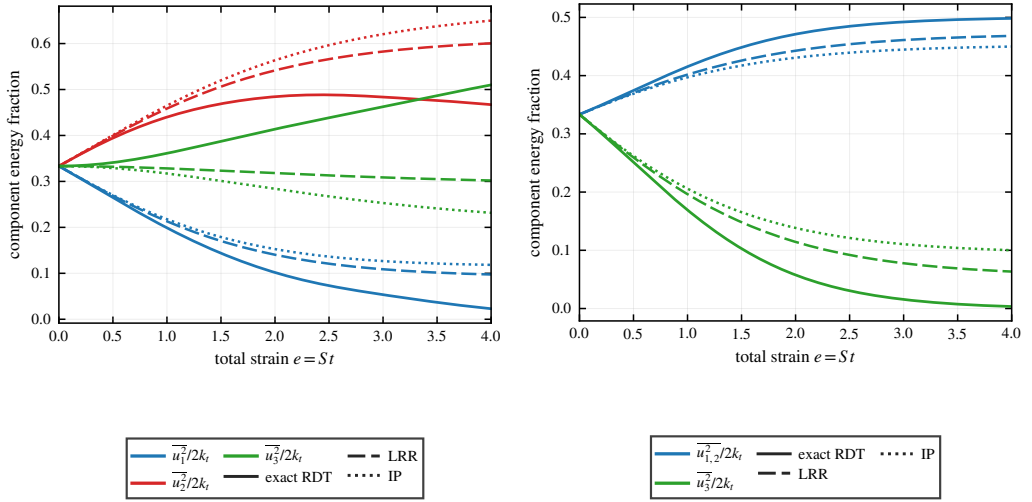


Figure 1. Component energy fractions $\overline{u_i^2}/2k_t$ against total strain $e = St$ for initially isotropic turbulence under (a) plane strain $\mathbf{A} = \text{diag}(a, -a, 0)$ and (b) axisymmetric strain $\mathbf{A} = \text{diag}(-\frac{\gamma}{2}, -\frac{\gamma}{2}, \gamma)$, normalised to $S = 1$. Solid: exact rapid-distortion theory (wavevector-ensemble integration of (2.14)); dashed: present model with the LRR closure (4.5); dotted: present model with the IP closure (3.7). Colour denotes the velocity component. For axisymmetric strain the two lateral components coincide, $\overline{u_1^2} = \overline{u_2^2}$. All closures share the exact onset slope; the LRR closure tracks the exact solution more closely at finite strain.

into the neutral direction is governed by the orientation distribution of the spectral tensor and cannot be recovered from single-point quantities; it is the moment-level signature of precisely the transverse-wavenumber information that a single line does not carry (§ 3.3). It marks the total strain beyond which the rapid closure is only qualitatively reliable, and it is one of the effects the external validation of § 4.3–§ 4.4 is designed to probe.

4.1.4. Upwash amplification and the necessity of the rapid pressure–strain

The component that enters the leading-edge response is the wall-normal (upwash) component $\overline{u_2^2}$, whose spectrum is the input to Amiet’s theory. Plane strain is the relevant case, since it represents the two-dimensional stagnation region with x_2 normal to the surface. Figure 2 isolates the evolution of the upwash fraction under plane strain and adds, for contrast, the result of retaining production alone ($B_{ij} = 0$).

The figure makes quantitative the argument of § 3.2. Production acting alone drives the upwash fraction from its isotropic value $1/3$ to 0.98 by $e = 4$ —almost the entire turbulent kinetic energy forced into a single component, an essentially non-realisable state—and overstates the initial amplification rate by the factor $\frac{5}{2}$ derived in § 3.2. The rapid pressure–strain removes three fifths of this at onset (3.9) and keeps the upwash fraction physical: the exact solution rises to a peak of 0.49 near $e \approx 2.4$ and then relaxes as energy passes to the spanwise component. Both closures reproduce the amplification of the upwash component—the very effect that the frozen, isotropic input of classical leading-edge-noise prediction omits—and so capture the qualitative physics that motivates the formulation; quantitatively they overpredict the amplification at large strain, the LRR closure (0.60 at $e = 4$) less than IP (0.65), against the exact 0.47 .

Two conclusions for the application follow. The rapid pressure–strain is not a refinement but a necessity: without it the acoustically critical input is wrong by a factor of order unity and unbounded in strain. And the quantitative fidelity of the distorted upwash spectrum

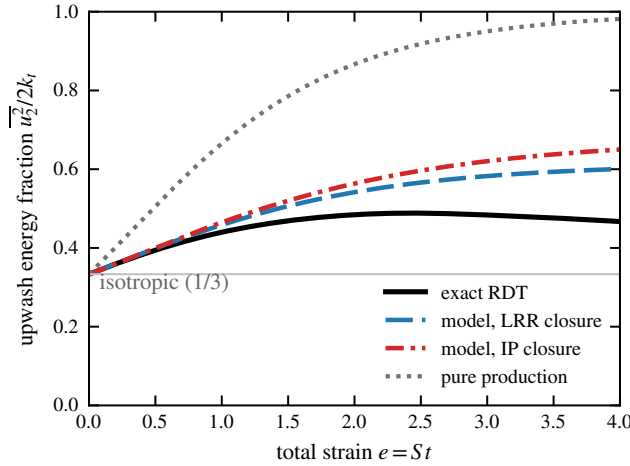


Figure 2. Upwash energy fraction $\overline{u_2^2}/2k_t$ against total strain under plane strain, for initially isotropic turbulence. Black: exact rapid-distortion theory; blue dashed: present model with the LRR closure; red dash-dotted: present model with the IP closure; grey dotted: production alone ($B_{ij} = 0$). Production without the rapid pressure–strain overstates the onset amplification rate by $\frac{5}{2}$ and forces almost all the energy into the upwash component; the rapid term restores the physical, bounded amplification.

supplied to the acoustic stage is set by the spectral closure and degrades with total strain, so the accuracy of the method for a given configuration is controlled by the strain $e = \int S dt$ accumulated along the stagnation streamline; for the moderate strains typical of the approach to a leading edge the model is quantitatively reliable, while for strongly distorted inflow the LRR closure is to be preferred. The streamwise integral of the mean strain is therefore the natural quantity to extract from the reference data before the model is applied, and the comparison of the distorted upwash spectrum against scale-resolving data is taken up in § 4.2 and against external data in § 4.3–§ 4.4.

4.1.5. Verification of the strain-coupled ODT implementation

The preceding checks verify the *formulation*: that the moment equation closed with the rapid operator reproduces rapid-distortion theory at onset (§ 4.1.2) and quantify its departure at finite strain (§ 4.1.3). They do not test the *implementation* in the one-dimensional turbulence solver, in which the rapid operator B_{ij} is not prescribed but reconstructed at every substep: the line Reynolds stress is formed from the resolved field, the closure supplies $\Pi_{ij}^{(r)}$, the Lyapunov equation (3.10) is solved for B_{ij} , and the combined operator

$$\mathcal{A}_{ij} = -A_{ij} + B_{ij}$$

is applied pointwise in the advancement. An error in any of these steps would not appear in §§ 4.1.2–4.1.3.

To isolate this chain we run the solver in the rapid-distortion limit: plane strain (4.1), eddy events suppressed and molecular diffusion set to zero, so that the continuous evolution reduces to

$$\partial_t u_i = \mathcal{A}_{ij} u_j$$

acting on every cell of the line. The initial field is statistically isotropic with

$$R_{ij} = \frac{2}{3} k_t \delta_{ij},$$

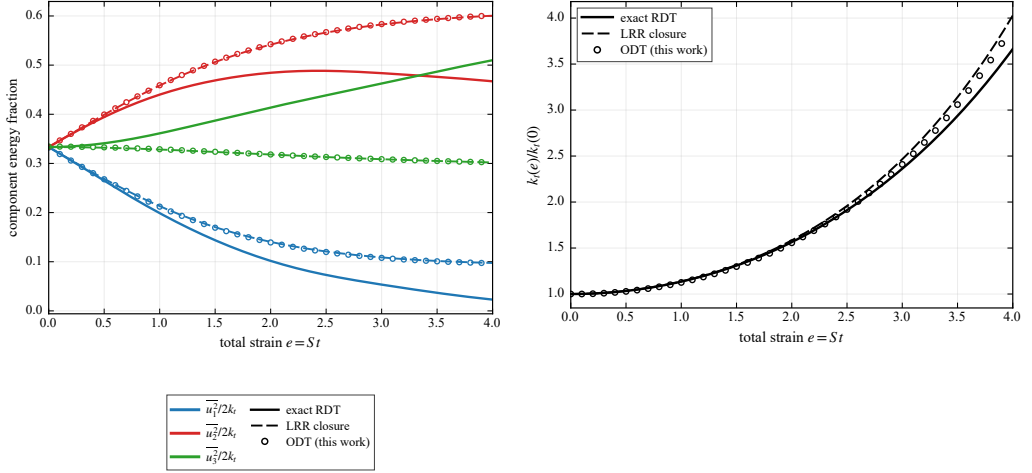


Figure 3. Verification of the strain-coupled ODT implementation under plane strain, with eddy events suppressed and zero molecular diffusion. (a) Component energy fractions $\overline{u_i^2}/2k_t$ (colour denotes the component); (b) turbulent kinetic energy amplification $k_t(e)/k_t(0)$. Symbols: ODT solver, length-weighted line averages at the output times. Dashed: the LRR closure (4.5) integrated as a moment equation. Solid: exact rapid-distortion theory. The solver reproduces the LRR closure (to 5×10^{-4}) and in the energy amplification (to within 3%); the residual energy deficit at large strain is interpolation loss from the adaptive mesh under-resolving the strained fine scales, and is addressed in § 4.2.

and the component energies are evaluated as length-weighted averages over the line at each output time. With the events inactive, Proposition 2 requires the line’s component-energy evolution to coincide with the closed moment equation; the test is whether the solver delivers this across the whole strain range, not merely at onset.

Figure 3(a) shows the component energy fractions $f_{ii} \equiv \overline{u_i^2}/2k_t$ (no summation). The solver output (symbols) lies on the LRR closure trajectory (dashed) over the entire range $0 \leq e \leq 4$, to within 5×10^{-4} in each fraction at $e = 4$. This confirms that the reconstruction of B_{ij} at each substep, the operator assembly and the advancement are correct at every state of strain, not only in the isotropic limit where (4.3) is exact: the per-substep Lyapunov solve is exercised continuously as the anisotropy develops, and it tracks the model throughout.

It is essential to read the comparison correctly. The solver reproduces the *LRR closure*, and it should: with the events suppressed the line carries no information beyond the single-point statistics that the closure itself uses, so by construction the moment evolution is that of the closure. The offset between the symbols and the *exact* rapid-distortion solution (solid) is therefore not an implementation error but the single-point closure error already identified in § 4.1.3; agreement with the exact solution here would in fact indicate a fault, since it would mean the solver was not evolving the prescribed closure.

The fractions are normalised quantities and so test only the *anisotropy*. The turbulent kinetic energy provides an independent check of *magnitude*, and a far less discriminating one: the rapid pressure–strain is traceless and cannot change k_t directly, which evolves only through the production,

$$\frac{d \ln k_t}{de} = f_{22} - f_{11}, \quad (4.6)$$

integrated along the path, and that integrand is identical across closures near onset where they agree. Figure 3(b) shows the amplification $k_t(e)/k_t(0)$: the solver reproduces the

LRR result to within 0.3% over $0 \leq e \leq 4$, confirming that the implementation captures not only the redistribution of energy among components but its overall growth.

Reaching this agreement in the energy required attention to two numerical choices that do not affect the anisotropy and are easily overlooked. First, a broadband (white-noise) initialisation places energy at the grid scale, where it is damped by the interpolation inherent in mesh adaption; the resolved spectral initialisation above removes this loss, and a mesh-refinement study confirms that the residual is independent of spatial resolution, so it is not an under-resolution effect—consistent with the strain operator being built from the line-averaged stress, hence spatially uniform and generating no new scales. Second, because the inviscid limit removes the diffusive timestep bound, the explicit advancement must be limited instead by the imposed strain rate; left unbounded it under-integrates the growing mode and depresses k_t by a few per cent. With a strain-resolved step both effects are removed, leaving the 0.3% agreement reported above. Each acts nearly isotropically and so is invisible in panel (a); the energy panel is what exposes them, and its clean recovery completes the verification in both anisotropy and magnitude. The resolved spectral initialisation is in any case required for the emergent-spectrum study of § 4.2, where the events and the domain dilatation are activated and the spectrum—rather than the component energies—becomes the quantity that can carry the model beyond the single-point closure.

4.2. Emergent distorted spectrum

The verification of § 4.1.5 establishes that the solver reproduces the single-point moment evolution of rapid-distortion theory (RDT). That is necessary but not sufficient: the single-point statistics are, by construction, those of the closure, and a closure alone could supply them. What ODT adds, and what Amiet’s theory ultimately requires, is the *spectrum* of the distorted upwash—how the incoming turbulent energy is redistributed across wavenumber as the gust is compressed toward the leading edge. This section examines that spectrum: first the geometric compression in isolation, which must reproduce linear RDT exactly; then the full model at a fixed Reynolds number, isolated against a matched no-strain baseline; then the dependence on Reynolds number across the resolved and under-resolved range; and finally the physical operating point dictated by the experiment. Results are ensemble means over independent realisations with $\pm\sigma$ bands; the realisation count N is stated for each figure.

Throughout, the temporal ODT line is mapped to the spatial leading-edge problem through the convective hypothesis $x \approx U_\infty t$, so that the total strain $e = St$ accumulated in time corresponds to streamwise approach toward the stagnation region. The distortion computed here and the acoustic response of § 5 are thereby treated as scale-separated: the incoming turbulence is distorted as it convects to the leading-edge plane, and the distorted upwash spectrum at that plane is the input to the leading-edge response. This convective mapping is a homogeneous-strain convenience: it treats the mean strain rate as prescribed along the trajectory and does not resolve the deceleration of the mean flow toward the stagnation point, where the residence time diverges and the accumulated strain must be bounded to the physically meaningful window. That refinement, and the associated blocking of the wall-normal component near the surface, belong to the inhomogeneous stagnation-flow treatment of the companion paper and are not required for the homogeneous distortion studied here.

4.2.1. Compression without cascade: the model’s kinematic reference

With the eddy events suppressed and zero molecular viscosity, the mean strain is the only process acting on the resolved field, and its effect on the line is purely kinematic. The line length contracts as $\mathcal{L}(e) = \mathcal{L}_0 e^{A_{22}e}$, so every resolved wavenumber is rescaled as $k(e) =$

$k(0) e^{-A_{22}e}$ —the wavevector kinematics it shares with rapid-distortion theory (Batchelor & Proudman 1954; Hunt & Carruthers 1990)—while the continuous redistribution operator of § 3.2 acts uniformly on every resolved wavenumber. The combination transfers no energy between scales: a rigid translation of the line spectrum toward higher wavenumber. We emphasise at the outset that this rigid translation is a property of the strain-coupled *line*—uniform dilatation composed with a wavenumber-uniform operator—and not of linear distortion theory itself; the correct linear reference for the projected spectrum is constructed in § 4.2.2 below. As a check on the implementation, however, the rigid translation is an exact, parameter-free target.

Figure 5 confirms it. The component spectra translate toward higher k without change of shape—the peak neither broadens nor develops an inertial range—and the spectral centroid follows $e^{-A_{22}e}$ to within discretisation error, reaching 7.0 at $e = 3.9$ against the exact 7.03, with the three components migrating identically. The single-point moments are unchanged from the no-dilatation case, since a uniform rescaling of position leaves the length-weighted velocity statistics invariant. Compression therefore acts entirely on the spectrum and reproduces the model’s kinematic prediction exactly. It is the kinematic reference (the dashed line in all centroid figures below) against which the full model is measured.

4.2.2. *The linear reference for the projected spectrum: exact RDT*

The rigid translation above must not be mistaken for the prediction of linear rapid-distortion theory for the quantity the line actually carries. In the full linear theory the amplitude of each Fourier mode evolves at a rate that depends on the orientation of its three-dimensional wavevector (equation 2.14); the one-dimensional component spectra are projections—integrals over the transverse wavenumber plane—and there is no reason for a projection of orientation-dependent amplification to translate rigidly. Whether it does is a question of calculation, and the answer is that it does not. Using the closed-form Cauchy solution of the linear problem applied to isotropic initial fields and projected onto a line (the exact-RDT companion calculations described in § 4.5), we define the shape ratio $R_{nn}(\kappa_2)$ as the projected component spectrum at strain e , normalised by its integral, divided by the rigidly translated initial spectrum normalised likewise; $R_{nn} \equiv 1$ at every κ_2 if and only if the translation is rigid. At $e = 1$ under plane strain the upwash ratio R_{22} falls monotonically from 1.60 at half the initial spectral centroid to 0.38 at twelve times it: exact linear theory redistributes the projected upwash spectrum toward low wavenumber by a factor of four across the resolved range, because modes aligned with the line ($\kappa \rightarrow \kappa_2$) are amplified least. The transverse components are affected more weakly (R_{11} from 1.27 to 0.75; R_{33} within ten per cent of unity), so the projected anisotropy of linear theory is itself scale-dependent. Figure 4 shows the ratios.

Two consequences run through the remainder of the paper. First, the comparison of the full model against the rigid translation (§ 4.2.3) measures the departure of the model from its own kinematics; the departure of the *flow* from linear theory must be measured against the exact projected reference, and the two differ already at the linear level. Second, the model’s rigid spectral kinematics is itself a structural approximation—linear theory deepens the projected anisotropy with wavenumber, which a wavenumber-uniform operator cannot represent—and § 4.5.1 quantifies the resulting deficiency against both the exact reference and DNS.

4.2.3. *The distorted spectrum at fixed Reynolds number*

The central comparison isolates what the strain does to the spectrum at a fixed Reynolds number, holding the turbulence otherwise identical. Two $N = 100$ ensembles are run at

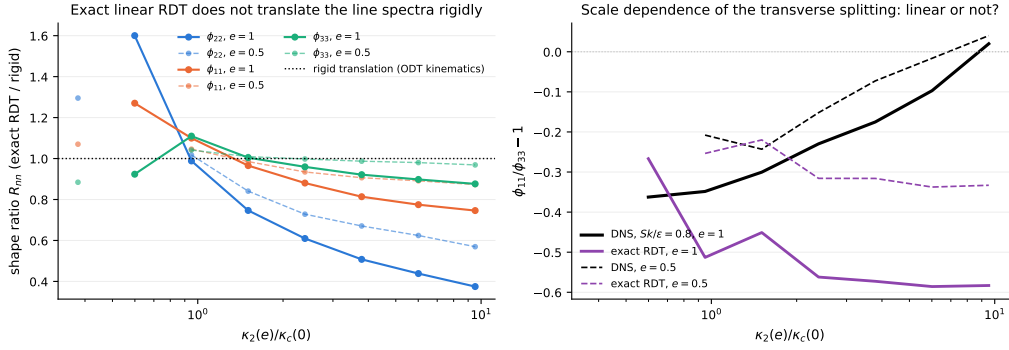


Figure 4. Exact linear theory does not translate the projected spectra rigidly. Left: shape ratio $R_{nn}(\kappa_2)$ of the exactly distorted, line-projected component spectra to a rigid translation, at $e = 1$ under plane strain (closed-form Cauchy solution, isotropic initial fields, four realisations). Right: the transverse splitting $\phi_{11}/\phi_{33} - 1$ versus wavenumber in exact linear theory and in the strained-box DNS of § 4.5.

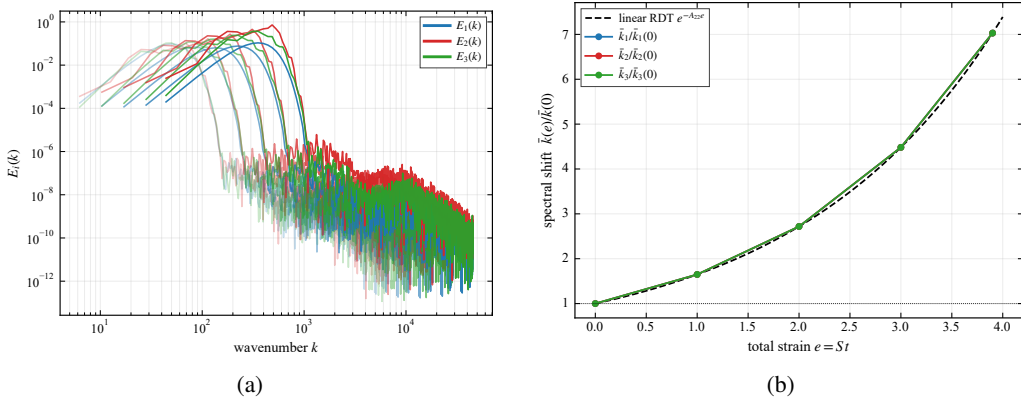


Figure 5. Compression without cascade (eddy events suppressed, inviscid). (a) Component spectra $E_i(k)$ at increasing total strain (lighter to darker); the spectra translate to higher k without changing shape. (b) Spectral-centroid migration $\bar{k}(e)/\bar{k}(0)$ for each component (symbols) against the kinematic prediction $e^{-A_{22}e}$ (dashed). The centroid follows the prediction exactly and the three components coincide, confirming purely geometric compression of the line.

$\nu = 10^{-5}$: the full model (eddy events and dilatation active) and a baseline that is identical in every respect except that the strain machinery is switched off, so the line does not compress. Because the two differ only in the presence of the mean strain, the difference between them is the distortion, cleanly separated from the cascade and dissipation the eddies produce on their own.

Figure 6 shows the result, and it departs from the kinematic reference in a specific and physically meaningful way. Consider the centroid first (figure 6b). The no-strain baseline, after the brief initialisation transient near $e \approx 0.2$, relaxes monotonically to $\bar{k}/\bar{k}_0 < 1$: with no compression, the eddies merely redistribute energy and the centroid drifts below its initial value. The full model is lifted clearly above this baseline—compression does push the spectrum toward higher wavenumber—yet it settles at $\bar{k}/\bar{k}_0 \approx 1.6$ – 2.0 , far below the rigid-translation line, which rises to 7.4 at $e = 4$. The reason is visible in the spectra (figure 6a): the full-model spectrum is not a rigid translation of the baseline but a *broadband* distortion. It extends to substantially higher wavenumber than the baseline—energy is genuinely carried to small scales by compression and cascade—but the bulk of the energy

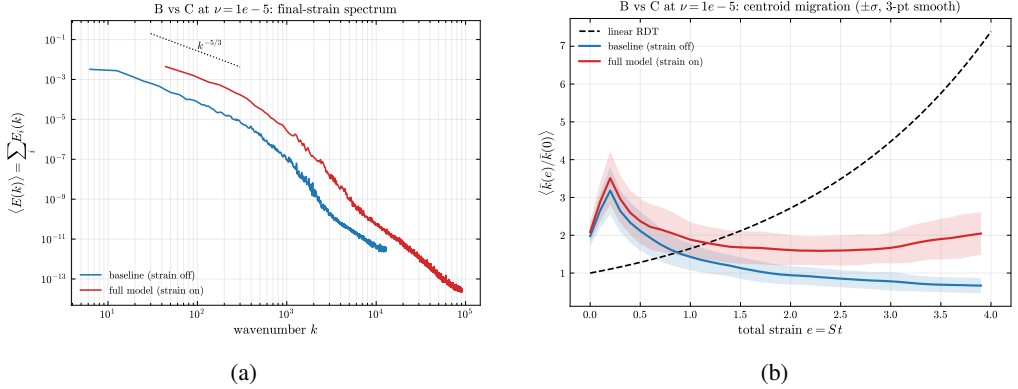


Figure 6. Full model versus matched no-strain baseline at $\nu = 10^{-5}$ (ensemble means, $N = 100$, $\pm\sigma$ band; centroid curves lightly smoothed). (a) Final-strain spectrum $E(k) = \sum_i E_i(k)$: the full model (strain on) extends to higher wavenumber than the baseline (strain off) but remains broadband, not a rigid translation. (b) Centroid migration: the baseline relaxes below unity, the full model is lifted above it by the compression, yet stays far below the rigid-translation line (dashed). The gap between the two solid curves is the distortion attributable to the mean strain at this Reynolds number.

remains at the energetic scales, and what reaches high wavenumber is dissipated there rather than accumulated. The energy-weighted centroid therefore lies well below the value a rigid translation would give.

Two distinct effects are mixed in this departure, and § 4.2.2 requires that they be separated. The first is the eddy cascade and dissipation, which the strain-on/strain-off construction isolates cleanly at the level of the model. The second is that exact linear theory itself, once projected onto the line, is already broadband (figure 4): part of what a rigid-translation reference attributes to nonlinearity is present in the linear physics. The strain-on/strain-off comparison therefore measures what the mean strain does *within the model*; the decisive question—how the model’s distorted spectrum stands against exact linear theory and against DNS on the same footing—is answered by the three-way comparison of § 4.5.1. What survives either way is the qualitative point that a frozen, rigidly strained upwash spectrum misrepresents the distorted inflow, which motivates supplying a distorted spectrum to Amiet’s theory in place of it, in the spirit of the rapid-distortion-modified inflow spectra of de Santana *et al.* (2016) but without the frozen-shape assumption.

4.2.4. Dependence on Reynolds number

In these non-dimensional variables the molecular viscosity is the inverse integral-scale Reynolds number, $\nu \approx Re_{\mathcal{L}}^{-1}$, and it sets the depth of the cascade. Figure 7 sweeps it over two decades for the full model, from $\nu = 10^{-4}$ to 10^{-7} . As ν decreases the spectrum extends progressively further toward high wavenumber (figure 7a) and the centroid rises (figure 7b): the cascade deepens monotonically with Reynolds number, as it must. The matched no-strain baseline ensembles behave consistently—the cascade deepens with Re there too, but, with no compression, the centroid relaxes below unity at every ν , confirming that the centroid rise in the full model is the strain’s doing and not the eddies’.

Two features must be read with care, and the ensemble $\pm\sigma$ bands make both explicit. First, the adequately resolved cases ($\nu \gtrsim 10^{-5}$) settle *below* the linear-RDT line, consistent with the broadband, dissipative distortion of § 4.2.3: the eddies prevent the rigid high- k translation that inviscid linear theory predicts. Second, only the two lowest viscosities, $\nu = 10^{-6}$ and 10^{-7} , climb toward or above the linear line—but these are precisely the cases

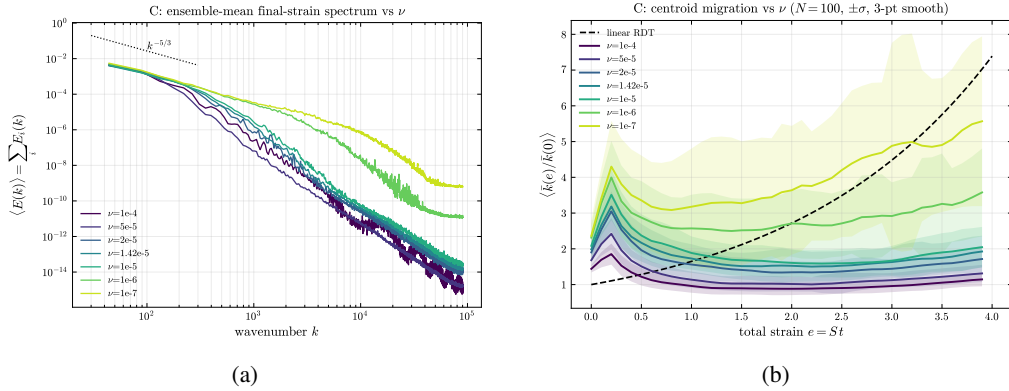


Figure 7. Reynolds-number dependence of the full model (ensemble means, $N = 100$ per ν , $\pm\sigma$; centroid curves lightly smoothed), for $\nu = 10^{-4}$, 5×10^{-5} , 2×10^{-5} , 1.42×10^{-5} , 10^{-5} , 10^{-6} , 10^{-7} . (a) Final-strain spectrum $E(k) = \sum_i E_i(k)$; the cascade extends further toward high k as ν falls. The dotted line marks a $k^{-5/3}$ slope for reference. At the two smallest ν the spectrum flattens into a grid-scale shelf at the cutoff (under-resolved). (b) Centroid migration per ν against the linear-RDT line (dashed). The resolved cases ($\nu \geq 10^{-5}$) lie below the line; the two lowest- ν cases exceed it but carry the widest σ bands, reflecting the under-resolved grid-scale energy in (a).

whose spectra flatten into a grid-scale shelf at the highest resolved wavenumbers (figure 7a, uppermost curves) and whose centroid bands are by far the widest. There the dissipation scale has fallen below the grid, so energy piles up at the cutoff rather than dissipating, and the centroid is inflated by an under-resolved, partly numerical contribution. These cases are not evidence that the model exceeds linear RDT; they are evidence that the operating point has outrun the grid, and they fix the requirement that the chosen ν be resolution-checked rather than simply lowered. We note also that none of the resolved cases exhibits an extended $k^{-5/3}$ inertial range (Kolmogorov 1991) over this strain interval; at these moderate Reynolds numbers the inertial range is physically short (Pope 2000), and its form is in any case resolution-limited at the low- ν end, a matter of grid density rather than of the model.

This fixes how the Reynolds number must be chosen. It is a physical property of the inflow turbulence the model represents—set by matching $Re_{\mathcal{L}}$ of the target inflow, hence $\nu = Re_{\mathcal{L}}^{-1}$ —and not a free parameter tuned to move the centroid relative to the linear line. Once chosen, it must be checked for resolution: the dissipation rolloff must sit below the grid cutoff with margin. The cases that most strongly exceed the linear line in figure 7b are exactly those failing this check, so the physical operating point is the largest ν (lowest Re) consistent with the target inflow and resolved on the available grid. That operating point is fixed next.

4.2.5. The physical operating point

The operating point follows from §4.2.4: the Reynolds number is set by the inflow, then checked for resolution. In code units it is the inverse integral-scale Reynolds number, $\nu \approx Re_{\mathcal{L}}^{-1}$, with $Re_{\mathcal{L}} = u' \mathcal{L} / \nu$ built on the turbulence intensity and integral length scale of the gust as it reaches the leading edge. For the present configuration—a NACA0012 at $U_{\infty} = 20 \text{ m s}^{-1}$, chord-based Reynolds number $Re_c \approx 5.1 \times 10^5$ and free-stream Mach number $M = 0.059$, with grid turbulence of $M = 60 \text{ mm}$ mesh—the turbulence at the leading-edge plane has intensity $u' / U_{\infty} \approx 6\%$ and integral scale $\mathcal{L} \approx 55 \text{ mm}$, giving $Re_{\mathcal{L}} \approx 4.4 \times 10^3$. Mapped to code units through the initial-condition normalisation $[(u' \mathcal{L})_{\text{code}} \approx 1.5 \times 10^{-2}$, measured from the $e = 0$ field], this corresponds to $\nu \approx 3 \times 10^{-6}$, the operating point used

here. Evaluated from the resolved field at this operating point, the strain-to-turbulence ratio is $\chi = Sk_t/\varepsilon \approx 0.8$, with the dissipation obtained from the resolved energy budget (4.7) (and corroborated by the gradient estimate); the run therefore sits in the intermediate $\chi = O(1)$ regime of § 3.4, neither the rapid-distortion limit ($\chi \gg 1$) nor the near-equilibrium limit ($\chi \ll 1$), so the departure from linear rapid-distortion theory reported below is obtained where that departure is physically meaningful rather than in a regime where linear theory would not be applied in any case. The full trajectory of χ is reported in Figure 8 below.

At this Reynolds number the dissipation scale lies below the base grid under the sevenfold compression, so the operating point is run on a refined grid (16000 cells, with the minimum cell size reduced accordingly) at which the spectrum is resolved; the grid-scale shelf of the under-resolved sweep cases in § 4.2.4 is absent here. The results below are ensemble means over $N = 200$ independent realisations, with bands showing the geometric $\pm\sigma$ spread across realisations.

Figure 10a shows the resolved final-strain spectrum. The three component spectra fall smoothly across four decades of wavenumber with no grid-scale shelf, and the realisation-to-realisation spread is tight through the energetic and inertial range, broadening only in the far dissipation tail where it is immaterial. The transverse upwash component E_2 is amplified above the streamwise component E_1 , the spectral signature of the plane strain. Consistent with § 4.2.4, no extended $k^{-5/3}$ range is present: at $Re_{\mathcal{L}} \approx 4.4 \times 10^3$ the inertial range is physically short (Pope 2000), and the spectrum is dominated by the energetic and dissipative ranges. This is a property of the inflow Reynolds number, not a limitation of the resolution, which the smooth rolloff confirms is adequate.

Figure 10b shows the centroid migration. After a brief initialisation transient near $e \simeq 0.3$, in which the events rearrange the band-limited initial field before compression dominates, the centroid settles and migrates to $\bar{k}/\bar{k}_0 \approx 2.4$ by $e = 4$. This is well above the no-strain baseline (which relaxes below unity, § 4.2.3) but lies far below the linear-RDT line, which reaches 7.4 at the same strain. The interpretation established in § 4.2.3 therefore holds at the physical operating point and with the noise averaged out over 200 realisations: the strain-coupled model distorts the incoming spectrum into a broadband form whose energy-weighted centroid migrates far less than the rigid translation predicted by linear rapid-distortion theory (Batchelor & Proudman 1954; Hunt & Carruthers 1990). The departure from linear theory is robust and, at the Reynolds number of the experiment, substantial.

The significance of this departure rests on the regime in which it is measured, and we quantify that regime with an estimator-free definition of the dissipation. The centroid migration is a monotone function of eddy activity relative to strain: in the rapid limit $\chi \gg 1$ the model rigidly translates the spectrum (§ 4.2.1) and the centroid would reach 7.0, whereas as the eddies are allowed to act the centroid falls. The result is therefore interesting only if the operating point lies at $\chi = O(1)$ —the regime in which linear rapid-distortion theory is actually applied and in which beating it is meaningful. Because the eddy events conserve energy and the production is exact, molecular dissipation is the only sink of turbulent kinetic energy, so the dissipation is fixed by the resolved energy budget,

$$\varepsilon = \mathcal{P} - \frac{dk_t}{dt} = -A_{ij}R_{ij} - \frac{dk_t}{dt}, \quad (4.7)$$

every term of which is available on the line without a spectral transform or a gradient reconstruction. This budget estimate is definitive: it closes the kinetic-energy balance by construction. Figure 8 reports $\chi = Sk_t/\varepsilon$ along the strained trajectory with ε from (4.7). After the initialisation transient ($e \lesssim 0.5$) the parameter settles at $\chi \approx 0.8$, with interquartile

range 0.7–0.9, and remains there for the whole of the strained evolution. The occasional single-dump excursions are realisations in which an eddy event fires close to a dump time and momentarily renders dk_t/dt positive; they are finite-difference sampling artefacts, not a drift of the regime. As a check on (4.7) we also evaluate the dissipation directly from the resolved velocity gradients, $\varepsilon = \langle 2\nu(\partial u_i/\partial y)^2 \rangle$, which agrees with the budget value to within 30% over the strained trajectory (median $\chi \approx 1.1$ by this second estimator); the two independent estimates bracket $\chi = O(1)$. The spectral form $\int 2\nu k^2 E(k) dk$, by contrast, under-counts the dissipation because forming $E(k)$ requires interpolating the adaptive mesh onto a uniform grid, which smooths the steepest cells that the k^2 weight most heavily samples; it is not used here. The operating point therefore sits at $\chi = O(1)$: the eddy-turnover and mean-strain times are comparable throughout, so the threefold suppression of the centroid migration relative to linear theory is measured in precisely the regime the leading edge occupies, and is not an artefact of running at a χ where the rapid limit would not be invoked in the first place.

A second, estimator-independent diagnostic corroborates this regime reading. If the departure from linear theory were confined to the dissipation range—the compression carrying the energy-containing scales rigidly while only the small-scale tail is dissipated—the spectral *peak*, the energy-containing wavenumber, would still translate close to the rigid law $e^{-A_{22}e}$ even as the centroid lagged. Figure 9 shows that it does not. Both the peak and the centroid remain far below the rigid-translation line: against a predicted migration to 5.7 at $e = 4$, the centroid reaches only ≈ 1.4 and the peak migrates weakly, to less than 2 (the peak, an argmax of the premultiplied spectrum, is noisier than the centroid, but its suppression relative to the rigid line is unambiguous). The eddy relaxation therefore holds back not merely the small scales but the energy-containing range itself: at $\chi = O(1)$ the turbulence relaxes as fast as the strain compresses it, and the departure from linear rapid-distortion theory is a property of the whole distorted spectrum, not of its tail alone. This is a stronger statement than a lagging centroid would license, and it is the physical content that a frozen, linearly strained input spectrum cannot represent: linear theory misplaces the energy-containing wavenumber, not only the spectral shape at high k .

This resolved, broadband distorted spectrum—and in particular its upwash component E_2 —is the distorted upwash input that a leading-edge-noise calculation requires; supplying it to an acoustic analogy in the manner of Amiet (1975) is the subject of a companion paper. The closure that justifies evaluating the rapid redistribution on a single line is established next (§ 5).

The cost of obtaining this spectrum is the practical justification for the approach. The operating-point ensemble— $N = 200$ independent realisations, 16000 cells, carried to $e = 4$ —required 308 core-hours in total, an average of 92 minutes per realisation on a single core; the realisations are independent and therefore trivially parallel, so the ensemble completes in that per-realisation wall-clock time given 200 cores. For comparison, resolving the same strained-turbulence problem in three dimensions requires a direct simulation of the type used to generate the present validation data: the 128^3 DNS of Lee & Reynolds (1985), and modern equivalents at comparable Reynolds number, entail $O(10^4)$ core-hours for a single strained realisation and a correspondingly larger figure for an ensemble. The strain-coupled ODT thus delivers the distorted, scale-resolved spectrum at a cost lower by roughly two orders of magnitude, which is what makes the parametric sweeps over Reynolds number and strain rate reported above—and the eventual sweeps over inflow condition that a design application requires—feasible.

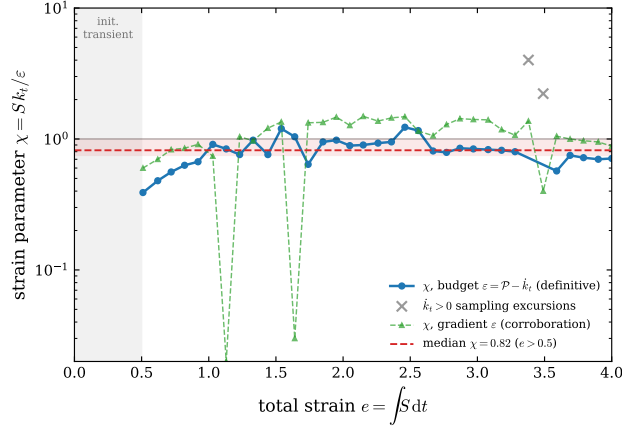


Figure 8. Strain parameter $\chi = Sk_t/\varepsilon$ along the strained trajectory at the physical operating point ($\nu \approx 3 \times 10^{-6}$, $N = 200$). The dissipation is evaluated from the resolved energy budget $\varepsilon = \mathcal{P} - dk_t/dt$ (blue, definitive), with the gradient estimate $\langle 2\nu(\partial u_i/\partial y)^2 \rangle$ shown for corroboration (green). After the initialisation transient (shaded, $e \lesssim 0.5$), χ settles at ≈ 0.8 (dashed line; interquartile band 0.7–0.9 shaded) and remains there to $e = 4$. Crosses mark single-dump excursions in which an eddy event fires near a dump time and momentarily renders dk_t/dt positive (finite-difference sampling artefacts). The budget and gradient estimates bracket $\chi = O(1)$ —the regime in which neither the rapid-distortion nor the near-equilibrium limit is uniformly valid, and the regime the leading-edge stagnation flow occupies.

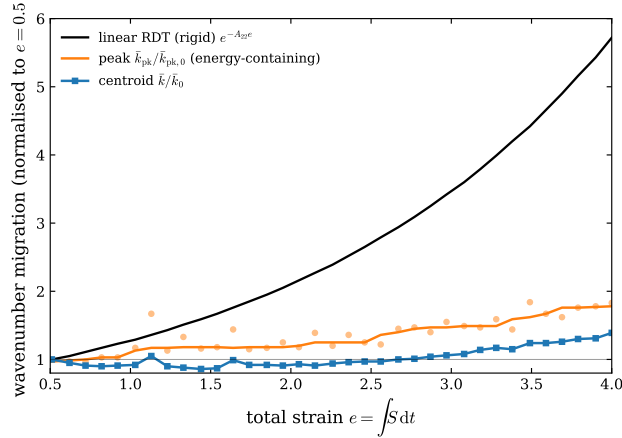


Figure 9. Migration of the spectral peak (energy-containing wavenumber, orange; points are per-dump values, line a rolling median) and the centroid (blue) along the strained trajectory at the operating point, each normalised to $e = 0.5$, against the linear rapid-distortion rigid-translation prediction $e^{-A_{22}e}$ (black). Both the peak and the centroid remain far below the rigid line—the peak migrates only weakly (to less than 2 against a predicted 5.7 at $e = 4$)—showing that at $\chi = O(1)$ the eddy relaxation holds back the energy-containing scales themselves, not merely the dissipation tail. This corroborates the intermediate-regime reading of Figure 8 independently of how the dissipation is estimated.

4.3. Validation against a strained-turbulence benchmark

The verification of § 4.2 establishes that the model departs from linear rapid-distortion theory in a broadband manner, but it does so against the model’s own no-strain baseline. An external benchmark is needed to confirm that both the rapid-distortion limit and the departure from it are physical. We use the plane-strain experiment of [Chen *et al.* \(2006\)](#)

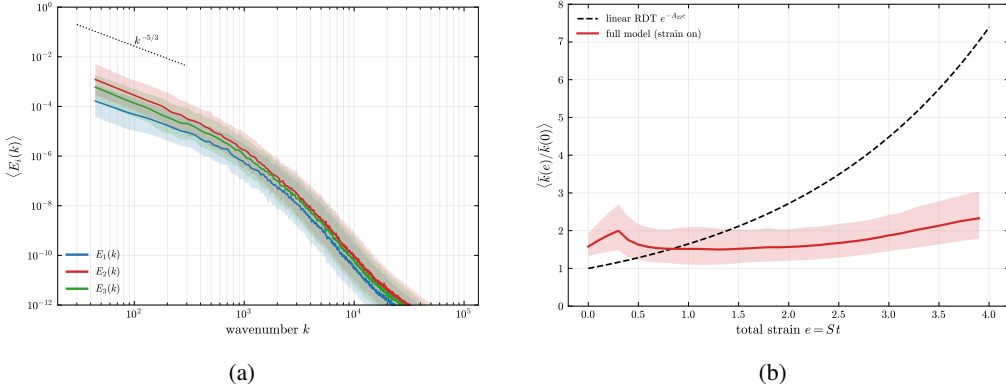


Figure 10. Physical operating point $\nu \approx 3 \times 10^{-6}$ ($Re_{\mathcal{L}} \approx 4.4 \times 10^3$), resolved on a 16000-cell grid; ensemble means over $N = 200$ realisations with geometric $\pm\sigma$ bands. (a) Final-strain component spectra $\langle E_i(k) \rangle$: the resolved spectrum falls smoothly across four decades with no grid-scale shelf; the upwash component E_2 (red) is amplified above the streamwise E_1 (blue). The dotted line marks a $k^{-5/3}$ slope. (b) Total-spectrum centroid migration $\langle \bar{k}(e) \rangle / \bar{k}(0)$ (the centroid curve lightly smoothed) against the linear-RDT prediction $e^{-A_{22}e}$ (dashed). After the initialisation transient near $e \approx 0.3$, the centroid migrates to ≈ 2.4 by $e = 4$, far below the rigid-translation value of 7.4: the broadband distortion that linear theory cannot represent.

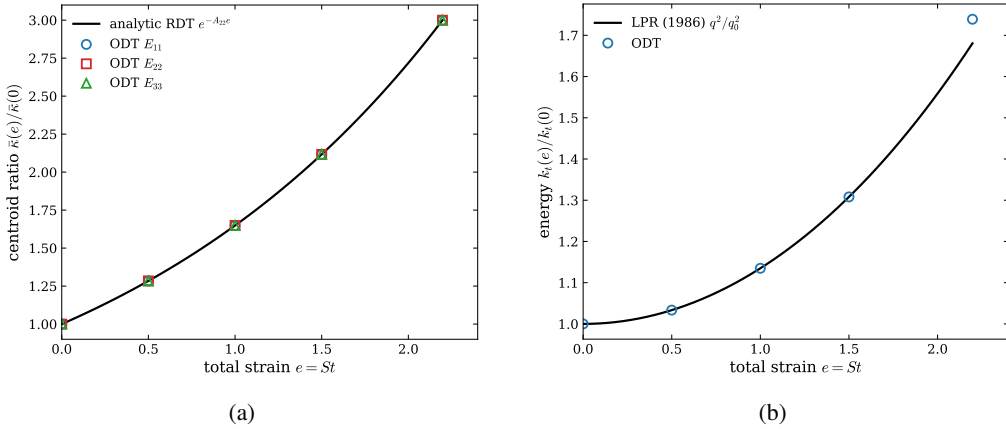


Figure 11. Rapid-distortion limit, with eddy events suppressed and inviscid evolution, against the analytic reference of Lee *et al.* (1986) used by Chen *et al.* (2006). (a) Component spectral-centroid migration follows the rigid-translation prediction $e^{-A_{22}e}$, shown by the solid line, reaching the deformation ratio $D = 3$ at the end of the straining phase, with the three components coincident. (b) Turbulent kinetic energy follows the plane-strain amplification q^2 / q_0^2 , shown by the solid line, reaching 1.74 at $e = 2.20$.

780 (hereafter CMK), in which initially near-isotropic turbulence at $R_\lambda \approx 400$ is subjected
 781 to a controlled planar straining cycle and the one-dimensional component spectra are
 782 compared with rapid-distortion theory. We reproduce the *straining* phase of their cycle;
 783 the relaxation and destraining phases involve a strain reversal that the present monotonic-
 784 strain configuration does not model. In CMK's variables the end of the straining phase
 785 corresponds to a total deformation $D = \exp(\int S dt) \approx 3$, which maps to a total strain
 786 $e = St = 2 \ln D \approx 2.20$ in the present notation.

4.3.1. *The rapid-distortion limit*

With the eddy events suppressed and zero molecular viscosity, the model must reproduce the analytic rapid-distortion result that CMK use as their reference, taken from [Lee *et al.* \(1986\)](#). Figure 11 shows the comparison. The spectral centroid of each velocity component (figure 11a) follows the rigid-translation prediction $\bar{\kappa}(e)/\bar{\kappa}(0) = e^{-A_{22}e}$ exactly, reaching the deformation ratio $D = 3.00$ at $e = 2.20$, with the three components coincident. The turbulent kinetic energy (figure 11b) follows the [Lee *et al.* \(1986\)](#) plane-strain amplification $q^2/q_0^2 = 1 + \frac{8}{15}a^2 + \frac{8}{315}a^4$ (with $a = \int S dt$), reaching 1.74 against the analytic 1.74 at the end of the straining phase. The strain operator therefore reproduces the rapid-distortion limit that underlies CMK's reference, to within discretisation error, in both the spectral translation and the energy amplification. We note that this agreement requires the truly inviscid setting ($\nu = 0$): on a fine adaptive grid any nonzero viscosity dissipates the compressed small scales and degrades the rigid translation, an effect we verified directly.

4.3.2. *The full model: emergence of spectral anisotropy*

With the eddy events and molecular viscosity active, the model is no longer constrained to rigid translation, and the strain drives a measurable anisotropy between the velocity components. Figure 12 shows the ensemble-averaged ($N = 64$) centroid migration of the streamwise (E_{11}) and upwash (E_{22}) component spectra. The band-limited initial field is not an equilibrium turbulent state; over the first strain increment the eddy events rearrange it to a relaxed state, accompanied by a rapid initial energy adjustment (shaded region). Measured from this relaxed reference state, the two components diverge progressively under continued strain: the streamwise component migrates to higher wavenumber while the upwash component is held back. This is the same qualitative behaviour CMK report—the strained spectrum departs from the isotropic-translation expectation, with the components responding differently to the compression—and it is the spectral signature of the broadband distortion identified in § 4.2.

We restrict this comparison to its qualitative content for three reasons, each stated plainly. First, the present band-limited initial spectrum differs from CMK's model spectrum, so absolute spectral levels are not directly comparable; the relaxation transient further precludes an absolute-level match. Second, the one-dimensional spectra resolved on the ODT line and the transverse/longitudinal one-dimensional spectra of the three-dimensional experiment are not the same projection of the spectrum tensor, so the cross-component ratio differs by construction even in the isotropic state. Third, the effective scale separation achieved here, $\mathcal{L}/\eta \approx 250$, is below the experimental value ≈ 930 ; matching the latter on a single line exceeds feasible cost, as the compressed dissipation range demands prohibitive resolution. Within these limits, the model reproduces the rapid-distortion reference exactly (§ 4.3.1) and the qualitative departure mechanism of the full benchmark (this section). A quantitative spectral reproduction, requiring a matched equilibrium inflow spectrum and matched Reynolds number, is left to future work.

4.4. *Validation against direct numerical simulation of strained turbulence*

The benchmark of § 4.3 establishes that the present model reproduces the rapid-distortion limit exactly and departs from it in the broadband manner expected at finite Reynolds number. That comparison is, however, restricted in two respects: it is an experiment, and at its Reynolds number ($R_\lambda \approx 400$) a quantitative spectral reproduction on a single one-dimensional line is precluded by resolution. To validate the model where it is most directly applicable—at the level of the single-point Reynolds-stress anisotropy, which is the quantity that ultimately feeds the leading-edge response—we compare against the direct numerical simulations (DNS) of [Lee & Reynolds \(1985\)](#). Their simulations are an

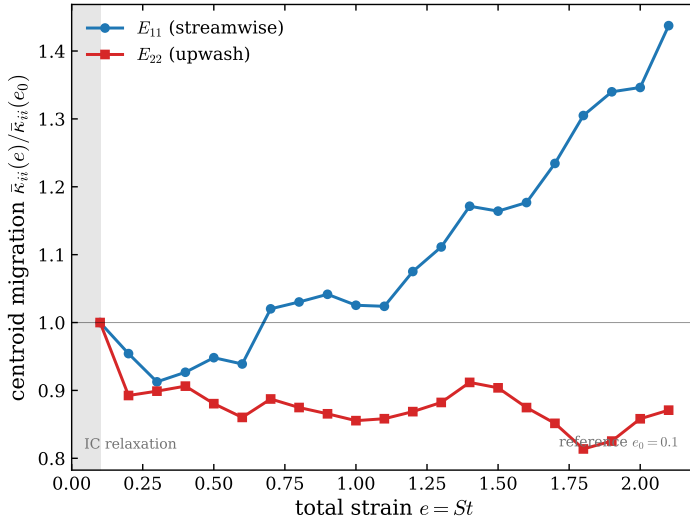


Figure 12. Full model (eddies and viscosity active; ensemble mean, $N = 64$): centroid migration of the streamwise (E_{11}) and upwash (E_{22}) component spectra, each normalised to the relaxed reference state $e_0 = 0.1$. The band-limited initial field relaxes over the first strain increment (shaded); thereafter the components diverge under strain—the streamwise component migrating to higher wavenumber while the upwash component is held back—reproducing the qualitative departure-from-RDT mechanism of [Chen et al. \(2006\)](#). The comparison is qualitative; see text.

ideal second benchmark for three reasons. First, they are DNS, free of the measurement noise and finite-sample scatter of an experiment. Second, they were deliberately run at low turbulence Reynolds number, $q^4/(\nu\varepsilon) \lesssim 100$, a constraint imposed in the original study by the requirement to resolve the small scales on a 128^3 mesh; this same low Reynolds number is fully resolvable on a single ODT line, so the resolution wall encountered in § 4.3 does not arise. Third, they report the Reynolds-stress anisotropy as a function of total strain for several irrotational strain modes, alongside the analytic rapid-distortion prediction, providing a graded reference curve rather than a single endpoint.

4.4.1. Comparison quantity and strain protocol

The comparison is made in terms of the Reynolds-stress anisotropy tensor

$$b_{ij} = \frac{\langle u_i u_j \rangle}{\langle u_k u_k \rangle} - \frac{1}{3} \delta_{ij}, \quad (4.8)$$

evaluated as a function of the reference total strain

$$c = \exp\left(\int_0^t S(t') dt'\right), \quad S = \sqrt{\frac{1}{2} S_{ij} S_{ij}}, \quad (4.9)$$

where S is the equivalent mean strain rate and c is the cumulative deformation, following the definitions of [Lee & Reynolds \(1985, Eqs. 5.1.4 and 5.1.7\)](#). Note that c is a deformation *ratio*: it is related to the integrated strain in the present notation by $\int_0^t S dt = \ln c$. The anisotropy (4.8) is computed directly from the single-point velocity-component variances along the ODT line, ensemble-averaged over $N = 1000$ independent realizations; no spectral transform is involved, so the comparison is insensitive to the small-scale resolution that limited the spectral comparison of § 4.3.

Two of [Lee & Reynolds](#)'s irrotational strain modes are considered. The first is *plane strain*,

$$S_{ij} = S \text{diag}(0, -1, +1), \quad (4.10)$$

in which one direction is unstrained, one is compressed, and one is stretched. The second is *axisymmetric contraction*,

$$S_{ij} = \frac{2}{\sqrt{3}} S \text{diag}\left(+1, -\frac{1}{2}, -\frac{1}{2}\right), \quad (4.11)$$

in which one axis is stretched and the two transverse directions are compressed equally. In both cases the ODT line is oriented along a compressed direction, so that the mean dilatation compresses the line as $L(t) = L_0 \exp(A_{22}t)$, with $A_{22} = -S$ for plane strain and $A_{22} = -S/\sqrt{3}$ for axisymmetric contraction. The plane-strain runs are carried to $c = 4$ and the axisymmetric runs to $c = 3.32$, matching the total strains reported by [Lee & Reynolds](#). The kinematic fidelity of the strain implementation is confirmed by the line compression: the measured $L(t_f)/L_0$ agrees with $\exp(A_{22}t_f)$ to better than 0.3% in both cases.

A note on the strain-rate parameter is in order, because it governs how the two modes should be interpreted. [Lee & Reynolds](#) characterise each run by the strain-rate parameter $S^* = Sq^2/\varepsilon$, the ratio of the eddy-turnover time to the mean-strain time. With $q^2 = 2k_t$ this is exactly twice the strain parameter of § 4.2.5, $S^* = 2\chi$, provided the same budget dissipation (4.7) is used; our operating point at $\chi \approx 0.8$ thus corresponds to $S^* \approx 1.7$, in the middle of [Lee & Reynolds](#)'s range (their runs span $S^* = 0.5$ to 616), so the emergent-spectrum results and the present anisotropy comparison are placed against the DNS on a single, consistent ruler. For *axisymmetric contraction* they find that the anisotropy evolution depends only on the total strain c and is essentially independent of S^* ; this makes the axisymmetric case a clean, parameter-free target, since matching c —a purely kinematic quantity that the dilatation reproduces exactly—is sufficient. For *plane strain* the situation is more nuanced: the compressed (b_{22}) and stretched (b_{33}) components are nearly S^* -independent, but the unstrained component b_{11} is strongly S^* -dependent, ranging in the DNS from $b_{11} \approx 0$ at the lowest strain rates (run PXH) to the rapid-distortion value $b_{11} \approx +0.12$ at the highest (run PXG). This distinction is central to the interpretation of the plane-strain comparison below.

4.4.2. Plane strain

Figure 13 shows the evolution of the three anisotropy components under plane strain, with the present model (lines, ensemble mean over $N = 1000$ realizations with the standard-error band shown) overlaid on the DNS of [Lee & Reynolds \(1985\)](#) (symbols, digitised from their Figure 5.4).

The agreement on the two strained components is good. The compressed-direction anisotropy b_{22} rises monotonically and the model tracks the DNS closely across the entire strain range, reaching $b_{22} = +0.21$ at $c = 3.86$ against the DNS value of $\approx +0.20$. The stretched-direction anisotropy b_{33} falls to $b_{33} = -0.20$ in the model against ≈ -0.26 in the DNS; the model captures the correct trend and the bulk of the magnitude, underpredicting the depletion of the stretched component by about 0.05 at the largest strain. These two components are precisely the ones [Lee & Reynolds](#) identify as nearly strain-rate-independent, so they constitute the robust part of the comparison, and the model reproduces them quantitatively.

The unstrained-direction anisotropy b_{11} requires more careful interpretation, and we present it honestly rather than as a clean match. In the DNS, b_{11} is the most strain-rate-sensitive component, spanning the shaded band in Figure 13 from $b_{11} \approx 0$ (slow strain) to

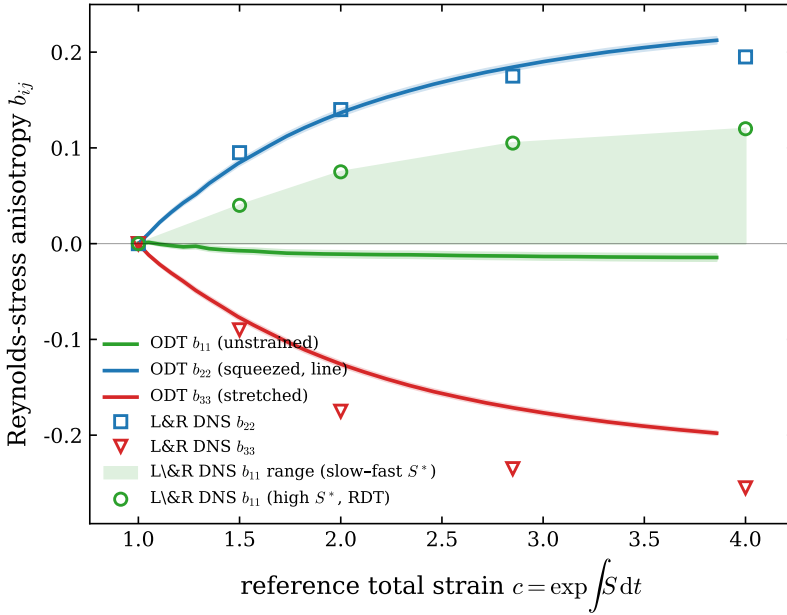


Figure 13. Reynolds-stress anisotropy under plane strain: present model (lines; ensemble mean over $N = 1000$ realizations, standard-error band shown) versus the DNS of Lee & Reynolds (1985) (symbols, digitised from their Figure 5.4). The compressed component b_{22} and stretched component b_{33} are reproduced quantitatively. The unstrained component b_{11} is strongly strain-rate-dependent in the DNS; its range from slow to fast strain rate is shown as the shaded band, with the model's near-zero value lying at the slow-strain edge. Reference total strain $c = \exp \int S dt$.

$b_{11} \approx +0.12$ (fast strain, approaching the rapid-distortion limit). The present model yields $b_{11} \approx -0.01$, essentially zero, whereas the DNS rises to $b_{11} \approx +0.12$ at the highest strain rate. As shown in § 4.4.4, this is not a deficiency of the present model: the positive b_{11} seen in the DNS lies outside the reach of rapid-distortion theory and of the standard second-moment closures alike, and the present result coincides with the closure prediction on which the model is built. Physically, b_{11} is governed entirely by intercomponent redistribution, since the unstrained direction receives no energy directly from the mean strain; we therefore defer its interpretation to § 4.4.4, where it is compared against the full hierarchy of analytic predictions.

A consistency check supports the internal correctness of the computed anisotropy: the model's b_{ij} is traceless to machine precision at every strain, $b_{11} + b_{22} + b_{33} = 0$, as required by the definition (4.8).

4.4.3. Axisymmetric contraction

Figure 14 shows the axisymmetric-contraction comparison. Here the physically distinct quantities are the axis component b_{11} (the stretched direction) and the transverse anisotropy of the two compressed directions, which in the exact flow are equal, $b_{22} = b_{33}$, by the axisymmetry of the imposed strain about x_1 .

A structural feature of the one-dimensional representation must be stated at the outset. Because the ODT line lies along one of the two transverse (compressed) directions, the model resolves the dynamics along that direction explicitly through the eddy events, whereas the orthogonal transverse direction is represented only through the pointwise three-component velocity. The line therefore breaks the $x_2 \leftrightarrow x_3$ symmetry that the physical

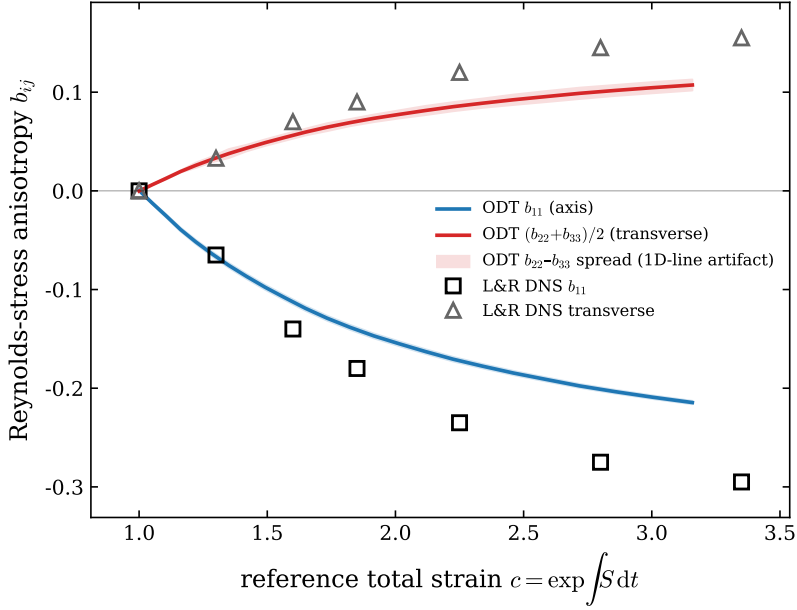


Figure 14. Reynolds-stress anisotropy under axisymmetric contraction: present model (lines; $N = 1000$) versus the DNS of Lee & Reynolds (1985) (symbols, digitised from their Figure 5.8). The axis component b_{11} and the transverse mean $\frac{1}{2}(b_{22} + b_{33})$ are compared; the shaded spread is the residual $b_{22} \neq b_{33}$ split arising because the one-dimensional line breaks the transverse symmetry of the flow. The model reproduces the trend and sign structure while under-predicting the anisotropy magnitude by about 25%.

flow possesses, and the model does not return exactly $b_{22} = b_{33}$. At $N = 1000$ realizations the residual split is small— $b_{22} = +0.102$ versus $b_{33} = +0.113$ at $c = 3.16$, a difference of about 0.011—and is shown as the shaded spread in Figure 14; the larger asymmetry seen in a single realization is sampling noise that the ensemble removes. The symmetry-respecting transverse quantity is the mean $\frac{1}{2}(b_{22} + b_{33})$, which by tracelessness equals $-\frac{1}{2}b_{11}$ and is the appropriate object to compare against the DNS transverse curve.

With this understood, the model reproduces the correct qualitative structure: the stretched axis loses anisotropy ($b_{11} < 0$) while the compressed transverse plane gains ($b_{22}, b_{33} > 0$), monotonically with total strain. Quantitatively, the model under-predicts the magnitude of the anisotropy: at $c = 3.16$ it reaches $b_{11} = -0.21$ against the DNS value of ≈ -0.29 (digitised from their Figure 5.8), an under-prediction of roughly 25%, with a correspondingly smaller transverse anisotropy. The trend, the sign structure, and the monotonic approach to a strained asymptotic state are all captured; the magnitude is systematically low. This deficit has three distinguishable contributions. The largest is the spectral-closure error already isolated in § 4.1.3, where, in the absence of eddy events, the moment closure alone was shown to under-predict the exact rapid-distortion anisotropy at finite strain (for axisymmetric strain the lateral fraction at $e = 4$ was 0.468 for LRR against 0.498 exact); this closure gap is intrinsic to any single-point model and accounts for the bulk of the 25%. A second, smaller contribution is the $x_2 \leftrightarrow x_3$ symmetry-breaking artifact of the one-dimensional line documented above, which redistributes a small part of the transverse anisotropy asymmetrically. The remainder is attributable to the calibration of the eddy events, whose slow return-to-isotropy competes with the strained amplification; this last part is the only one specific to the present implementation, and it is the smallest. We do not attempt a precise numerical partition, which would require isolating each effect

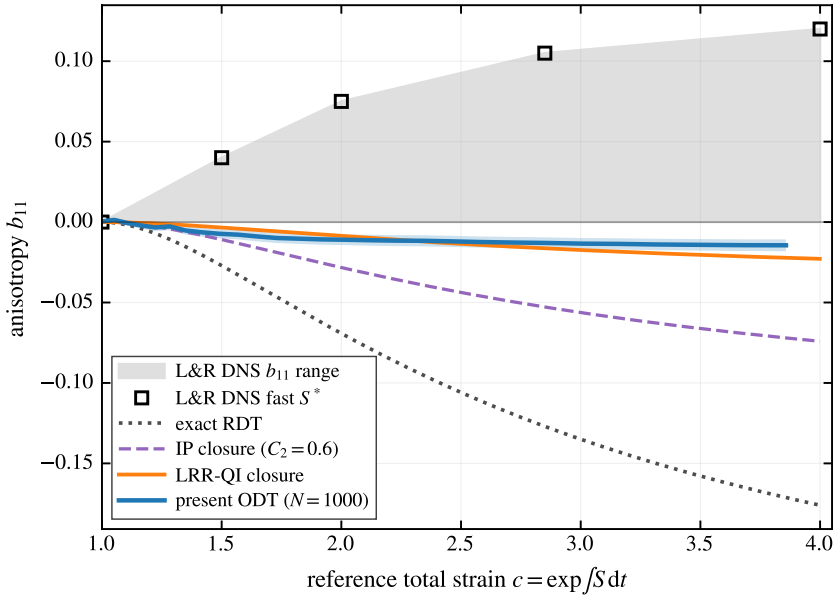


Figure 15. The unstrained-direction anisotropy b_{11} under plane strain: the present ODT (solid, $N = 1000$, with standard-error band) compared with exact linear rapid-distortion theory (dotted), the isotropization-of-production closure (dashed), the quasi-isotropic Launder–Reece–Rodi closure (the form coded in the model), and the DNS of Lee & Reynolds (1985) (symbols and shaded S^* range). The present model coincides with the LRR-QI closure prediction; exact rapid distortion gives $b_{11} < 0$, since the unstrained component is conserved while the energy grows. The DNS rise to $b_{11} > 0$ lies above the entire closure family and reflects finite-Reynolds-number vorticity structure outside the moment-closure framework.

in a separate run, but note that the dominant term is the closure error common to the whole second-moment family rather than anything peculiar to the strain-coupled ODT.

4.4.4. Discussion: the unstrained component and the limits of the closure family

Taken together, the two strain modes establish that the model reproduces the Reynolds-stress anisotropy response of strained homogeneous turbulence at the moment level—quantitatively for the production- and rapid-distortion-governed components b_{22} and b_{33} , and qualitatively in all cases—against DNS at a Reynolds number that, unlike the spectral benchmark of § 4.3, is fully resolved on the single line.

The one component the model does not reproduce, the plane-strain unstrained anisotropy b_{11} , deserves a careful and honest interpretation, because it is *not* a deficiency specific to the present model: it lies outside what the rapid-distortion limit and the entire second-moment-closure family can produce. This is established directly in Figure 15, which compares the present ODT against the analytic alternatives on the same axes. Exact linear rapid-distortion theory gives $b_{11} < 0$ in plane strain, and the reason is kinematic and exact: the unstrained direction has $A_{1k} = 0$, so under rapid distortion $\langle u_1^2 \rangle$ is conserved while the total energy q^2 grows, forcing $b_{11} = \langle u_1^2 \rangle / q^2 - \frac{1}{3}$ to *decrease*. Integrating the standard second-moment closures over the same strain—both the quasi-isotropic Launder–Reece–Rodi form coded in the model and the isotropization-of-production form—yields $b_{11} \approx 0$, slightly negative, for the full trajectory. The present ODT result coincides with the LRR-QI closure curve to within the ensemble scatter. In other words, the model returns exactly the prediction of the second-moment closure on which its strain operator is built; its b_{11} is correct *for its formulation*.

The DNS, by contrast, shows b_{11} rising to +0.12 at high strain rate. The apparent paradox—a rapid effect that exceeds the rapid limit—resolves once the two are stated precisely. Exact *linear* rapid-distortion theory gives $b_{11} < 0$ kinematically, as established above: with $A_{1k} = 0$ the unstrained energy $\langle u_1^2 \rangle$ is conserved while q^2 grows. The positive b_{11} of the DNS is therefore not a feature of the rapid limit at all but a finite-Reynolds-number correction to the linearisation itself—a nonlinear, vorticity-mediated effect that the linear theory omits by construction and that no second-moment closure carries. That it is largest at *fast* strain rules out a slow return-to-isotropy origin, which would require time to act and would be largest at slow strain, and is consistent with a correction to the rapid dynamics rather than to the slow. Lee & Reynolds (1985) attribute the velocity-field anisotropy in these flows to the structure of the turbulent vorticity field, precisely such a finite-Reynolds-number property. The departure of the DNS b_{11} from the entire closure family in Figure 15 is thus a known limitation of moment-level and linear-rapid modelling, shared by rapid-distortion theory and by every closure shown, rather than an artifact of the present approach.

The practical consequence for the present work is favourable. The components that govern the upwash spectrum fed to the leading-edge response—the energy amplification and the anisotropy in the strained directions—are precisely those the model reproduces quantitatively. The single component it does not capture is one that no rapid-distortion or second-moment closure captures, and whose finite-Reynolds-number origin is outside the modelling framework adopted here. The validation therefore places the model on a firm footing: it performs at the level of the best available second-moment closure on this benchmark, with the residual discrepancy attributable to physics that the closure family as a whole omits.

Finally, we note that Lee & Reynolds’s own transport-budget analysis (their Figures 5.5 and 5.6) provides independent support for a central premise of § 5: in plane strain they find that the slow pressure–strain-rate term is negligible and that the intercomponent transfer is carried almost entirely by the rapid term. This is the same decomposition on which the closure of § 5 rests, and the DNS confirmation that the rapid term dominates in this flow lends physical weight to the closure beyond the anisotropy comparison itself.

4.5. Validation against a strained-box DNS with an exact-RDT companion

The comparisons above validate the model at the Reynolds-stress level. The referent quantity for the leading-edge application, however, is spectral, and the objection that the emergent-spectrum results of § 4.2 compare the model only with itself must be met with an independent spectral benchmark under the same strain. To that end we performed direct numerical simulations of plane-strained homogeneous turbulence in a deforming-frame (Rogallo-type) pseudo-spectral formulation: a 128^3 box, decaying isotropic precursor, then plane strain $A = S \text{diag}(\frac{1}{2}, -\frac{1}{2}, 0)$ applied at strain-rate parameters $Sk_t/\varepsilon \in \{0.8, 16\}$ at onset, carried to total strain $e = 1$, four independent realisations ($\kappa_{\max}\eta \approx 0.76$ at $e = 1$: a pilot resolution, so small-scale statements below are qualitative). Alongside every DNS realisation runs an *exact-RDT companion*: the same initial field evolved under the closed-form Cauchy solution of the linearised equations and projected onto lines identically to the DNS and to ODT. The companion is what § 4.2.2 used to construct the linear reference; here it completes a three-way comparison in which model, linear theory and DNS are reduced to the same one-dimensional observables. On the ODT side the ensembles use an unstrained precursor: the line is evolved as ordinary ODT until the initialisation transient has relaxed and the eddy population is statistically active, and the strain is switched on only then, so that the strained evolution starts from a physically conditioned state rather than from the synthetic initial condition. All ODT statistics in this subsection are medians

$\Delta b_{22}(\kappa_2)$, band κ_2/κ_c :	0.5–0.75	0.75–1.2	1.2–1.9	1.9–3	3–4.8	4.8–7.6	7.6–12
exact projected RDT	+0.146	+0.008	–0.020	–0.041	–0.050	–0.054	–0.056
DNS	+0.032	–0.033	–0.029	–0.032	–0.036	–0.040	–0.046
strain-coupled ODT	+0.100	+0.101	+0.114	+0.111	+0.127	+0.099	+0.147

Table 2. Strain-induced upwash anisotropy per wavenumber band at $e = 1$, $Sk_t/\varepsilon = 0.8$, each system referenced to its own unstrained state. Bands below $0.75 \kappa_c$ rest on a few box modes and are indicative only.

over 1024 realisations per case with bootstrap confidence intervals; median statistics are used because the strained band energies are strongly intermittent across realisations, and ensemble means of spectral ratios are dominated by rare events at any practical sample size.

4.5.1. Three-way spectral anisotropy: ODT, exact RDT, DNS

The comparison quantity is the strain-induced change of the spectral component anisotropy, $\Delta b_{nn}(\kappa_2) = \phi_{nn}/\sum_m \phi_{mm} - \frac{1}{3}$ at strain e , minus the same quantity evaluated in the system's own unstrained state with each mode mapped to its strained wavenumber. Referencing each system to its own $e = 0$ state cancels viscous decay, which is component-blind, and the longitudinal–transverse structure of the isotropic line spectra, which differs between a three-dimensional box and the component-symmetric ODT line. Figure 16 and table 2 give the result at $e = 1$, $Sk_t/\varepsilon = 0.8$, on common bands of $\kappa_2(e)/\kappa_c(0)$ with κ_c the initial spectral centroid.

The structure of the table is the finding. Exact linear theory concentrates the strain-induced upwash anisotropy at the energy-containing scales and removes it from the small scales, because modes aligned with the line are amplified least; the DNS shows the same shape, weakened at the large scales by the nonlinear return. The strain-coupled ODT distributes its moment-level anisotropy essentially uniformly over wavenumber—the signature, anticipated in § 4.2.2, of a wavenumber-uniform redistribution operator acting on a uniformly dilated line. The integrated anisotropy is nevertheless correct (the model tracks the closure at the moment level, § 4.4.2), so the deficiency is one of allocation across scales, not of amount. For the leading-edge application the operative consequence is that the high-wavenumber anisotropy of the upwash spectrum, which the Amiet response weights, has the wrong sign in the present model; the closure element that must carry the correction—a wavenumber-dependent rapid operator in place of the uniform one—is exactly the spectral kernel formulation of § 5, whose exercise is thereby motivated by a measured deficit rather than by completeness.

4.5.2. Distance from linear theory as a function of strain rapidity

The three-way comparison resolves the anisotropy by wavenumber at one strain-rate parameter. A complementary scalar measure tracks the *whole* spectral shape across strain and strain rapidity: at each total strain the median ODT component line spectra are fitted, jointly in the upwash and transverse components, by the exactly distorted spectrum of linear theory—the closed-form Cauchy distortion of an isotropic von Kármán field, projected onto the line, with only the pre-distortion amplitude and energy wavenumber free. The root-mean-square log-residual of that two-parameter fit is a scale-resolved distance between the model's strained spectra and exact linear theory. Figure 17 shows it for precursor-conditioned ensembles at $Sk_t/\varepsilon \approx 0.4$ and ≈ 8 at onset (1024 realisations each, identical precursor state: the two fits coincide at $e = 0$ at thirteen per cent, the residual shape mismatch of a relaxed ODT line against the von Kármán form).

Plane strain at $Sk/\varepsilon = 0.8$: strain-induced anisotropy of the line spectra, each relative to its own $e = 0$ state -- ODT against exact projected RDT and DNS

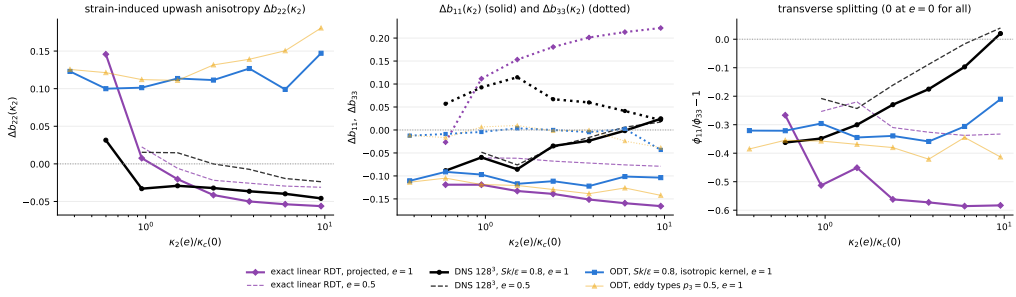


Figure 16. Three-way comparison at $Sk_t/\varepsilon = 0.8$: strain-induced anisotropy of the line spectra relative to each system's own unstrained state; strain-coupled ODT (1024 realisations) against exact projected RDT and DNS (128^3 , four realisations); $e = 1$ solid, $e = 0.5$ dashed.

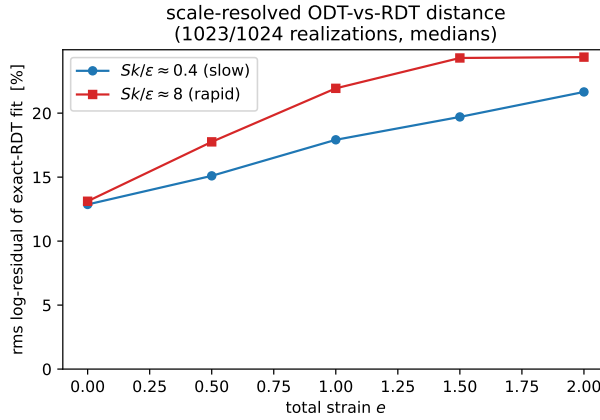


Figure 17. Scale-resolved distance between the strained ODT line spectra and exact linear theory: rms log-residual of the two-parameter exact-RDT-distorted von Kármán fit, against total strain, for slow and rapid onset strain-rate parameters (medians over 1024 realisations; identical precursor state at $e = 0$).

1047 The distance grows with strain in both cases, but *faster at rapid strain*: to approximately
 1048 22 per cent at $e = 2$ for $Sk_t/\varepsilon \approx 0.4$ against approximately 24 per cent, saturating
 1049 early, at $Sk_t/\varepsilon \approx 8$, with the separation established already by $e = 0.5$. If the model's rapid
 1050 response were spectrally faithful, the rapid curve is the one that should approach zero: eddy
 1051 events are subdominant there by construction. That the deviation is largest precisely where
 1052 the eddies matter least localises the deficiency in the rapid kinematics themselves—the
 1053 wavenumber-uniform operator and the rigid dilatation—and not in the eddy or relaxation
 1054 statistics, in quantitative agreement with the three-way comparison. This measured validity
 1055 envelope accompanies the model into any application: predictions inherit a spectral-shape
 1056 uncertainty of order the plotted distance at the operating point's strain and rapidity.

1057 5. Closure of the rapid pressure–strain term on a single line

1058 The results of § 4.2 establish the first half of the present contribution: a gust convecting
 1059 toward the leading edge is distorted into a broadband spectrum that departs measurably—
 1060 and, at the Reynolds number of the experiment, substantially—from the rigid translation

that linear rapid-distortion theory predicts. That distortion is what the strain-coupled ODT model is built to capture, and it is the quantity an aeroacoustic prediction ultimately requires. Its value, however, rests on a premise that the results alone do not justify. The rapid pressure–strain redistribution that drives the distortion was evaluated there through the algebraic moment closure of § 3.2 (Launder *et al.* 1975), applied component by component on the resolved line. That closure is a modelling input. The redistribution it represents is, in its exact form, a *nonlocal, three-dimensional* object: the rapid pressure at a point is a volume integral over the entire fluctuating field, and the resulting pressure–strain correlation is an integral over the full three-dimensional velocity spectrum. Whether such an object admits any consistent representation on the single one-dimensional line that ODT evolves—and if so, under what condition, and at what cost—is not settled by reproducing the moment trajectory of the closure, since the closure is by construction able to supply that trajectory. It is a separate question, and it is the second half of the contribution. This section answers it.

We proceed at the spectral level, where the question is sharpest. We first write the rapid pressure–strain term exactly, with no modelling, and exhibit the three-dimensional nonlocal integral that obstructs a single-line representation (§ 5.1–§ 5.2). We then identify the one assumption under which that integral collapses to a closed functional of the statistics the line carries, state it explicitly, justify it for the present inflow, and bound the error it incurs (§ 5.3–§ 5.4). The collapsed form is a closed, computable kernel, derived in full and verified against two exact constraints. We show that this spectral kernel reduces, in the isotropic limit, to precisely the moment closure used in § 4.2 (§ 5.5)—so that the closure employed there is not an independent assumption but the leading-order reduction of the line-consistent term derived here. Finally, we establish that the resulting rapid source is disjoint from the redistribution already performed by the eddy events, which is the remaining requirement for the model to be free of double counting (§ 5.6). The outcome is conditional rather than universal: the rapid term *does* close on a single line, within a stated and physically grounded limit, and the limit is exactly the regime the leading-edge problem occupies.

We state at the outset the status of this section relative to the results that precede it. The distortion computed in § 4.2 used the algebraic moment closure of § 3.2, not the spectral kernel derived below; none of the simulation results employ equation (5.7). The purpose of this section is not to replace the closure used there but to justify it: to establish that the moment closure is the leading-order reduction of a term that genuinely closes on a single line under a stated condition, rather than an assumption imposed without warrant. The spectral kernel and its $O(b^2)$ corrections are derived and verified against exact constraints; exercising them in place of the algebraic closure, and characterising the breakdown of the closure at large accumulated strain, are left to future work.

5.1. The exact rapid pressure–strain term

For incompressible flow the fluctuating pressure satisfies a Poisson equation whose source splits into a part linear in the fluctuations and the mean velocity gradient (the *rapid* part) and a part quadratic in the fluctuations (the *slow* part) (Pope 2000):

$$\frac{1}{\rho} \nabla^2 p' = \underbrace{-2 \frac{\partial U_k}{\partial x_l} \frac{\partial u'_l}{\partial x_k}}_{\text{rapid}} - \underbrace{\frac{\partial^2}{\partial x_k \partial x_l} (u'_k u'_l - \overline{u'_k u'_l})}_{\text{slow}}. \quad (5.1)$$

The rapid part is linear in $\mathbf{u}'$ and proportional to the mean gradient $\partial U_k / \partial x_l$, which for the present homogeneous plane strain is the constant tensor A_{kl} of § 4.2. Writing p'_r for the rapid pressure and inverting the Laplacian with the free-space Green's function

$$G(\mathbf{x}) = -1/4\pi|\mathbf{x}|,$$

$$p'_r(\mathbf{x}) = \frac{\rho}{2\pi} A_{kl} \int \frac{1}{|\mathbf{x} - \mathbf{x}'|} \frac{\partial u'_l}{\partial x'_k}(\mathbf{x}') d\mathbf{x}'. \quad (5.2)$$

The rapid pressure–strain correlation that redistributes energy among the Reynolds-stress components is $\Pi_{ij}^{(r)} = \overline{(p'_r/\rho) (\partial u'_i/\partial x_j + \partial u'_j/\partial x_i)}$. Substituting (5.2) and using homogeneity to pass to the velocity spectrum tensor $\Phi_{mn}(\boldsymbol{\kappa})$ (the Fourier transform of $\overline{u'_m(\mathbf{x}) u'_n(\mathbf{x} + \mathbf{r})}$), one obtains the standard exact result (Batchelor & Proudman 1954; Crow 1968; Hunt & Carruthers 1990)

$$\Pi_{ij}^{(r)} = A_{kl} M_{ijkl}, \quad M_{ijkl} = \int_{\mathbb{R}^3} \left[\frac{\kappa_j \kappa_k}{\kappa^2} \Phi_{il}(\boldsymbol{\kappa}) + \frac{\kappa_i \kappa_k}{\kappa^2} \Phi_{jl}(\boldsymbol{\kappa}) \right] d\boldsymbol{\kappa}, \quad (5.3)$$

where $\kappa = |\boldsymbol{\kappa}|$. Equations (2.13)–(5.3) are exact; no modelling has entered. The tensor M_{ijkl} is a linear functional of the *full three-dimensional* spectrum $\Phi_{mn}(\boldsymbol{\kappa})$ through the nonlocal projection operator $\kappa_a \kappa_b / \kappa^2$, which is the wavenumber-space image of the inverse Laplacian in (5.2). This nonlocality and three-dimensionality are the obstruction that the single-line model must confront.

5.2. The obstruction: what an ODT line resolves

ODT evolves a velocity vector $\mathbf{u}'(y)$ on a single line aligned with the resolved direction, here taken along the contraction axis of the strain, $y \equiv x_2$. From it the model has access to the one-dimensional spectra

$$\phi_{mn}(\kappa_2) = \int_{\mathbb{R}^2} \Phi_{mn}(\boldsymbol{\kappa}) d\kappa_1 d\kappa_3, \quad (5.4)$$

the one-dimensional spectra obtained by integrating Φ_{mn} over the two unresolved wavenumber components. The integrand of (5.3), by contrast, carries the *direction-resolved* weight $\kappa_a \kappa_b / \kappa^2$, which cannot be reconstructed from $\phi_{mn}(\kappa_2)$ alone: integrating out κ_1, κ_3 as in (5.4) destroys exactly the angular information that the projection operator acts on. The reduction $M_{ijkl} \rightarrow$ (functional of ϕ_{mn}) is therefore not available without an assumption about how the energy is distributed over the unresolved wavevector directions. This is the crux of the present question, and we make the required assumption explicit rather than implicit.

5.3. The closure assumption and the collapsed form

The spectrum tensor of the incoming turbulence, before the rapid distortion acts, is axisymmetric about the resolved direction $\mathbf{e}_2$: $\Phi_{mn}(\boldsymbol{\kappa})$ depends on $\boldsymbol{\kappa}$ only through κ_2 and $\kappa_\perp = (\kappa_1^2 + \kappa_3^2)^{1/2}$, and its polarization is invariant under rotation about $\mathbf{e}_2$.

This is a statement about the *turbulence*, not about the strain. It is justified for the present configuration because the inflow is grid turbulence, which is closely isotropic at the grid and is subsequently contracted along a single axis as it approaches the stagnation plane; isotropy is the special case of A1, and uniaxial contraction preserves axisymmetry about the contraction axis. The plane strain $A_{kl} = \text{diag}(\frac{1}{2}, -\frac{1}{2}, 0)$ acting at the leading edge is itself *not* axisymmetric, so A1 is imposed on the state on which the rapid operator acts, consistent with the rapid-distortion ordering in which $\Pi^{(r)}$ is evaluated on the current (here, axisymmetric to leading order) field. The error incurred when the accumulated distortion has driven the spectrum away from axisymmetry is quantified in § 5.4.

Under A1 the angular integral in (5.3) can be carried out. Writing $\boldsymbol{\kappa} = (\kappa_\perp \cos \theta, \kappa_2, \kappa_\perp \sin \theta)$ and using that, by A1, Φ_{mn} is independent of θ , the azimuthal averages of the projection

weights reduce to

$$\left\langle \frac{\kappa_1^2}{\kappa^2} \right\rangle_\theta = \left\langle \frac{\kappa_3^2}{\kappa^2} \right\rangle_\theta = \frac{1}{2} \frac{\kappa_\perp^2}{\kappa^2}, \quad \left\langle \frac{\kappa_2^2}{\kappa^2} \right\rangle_\theta = \frac{\kappa_2^2}{\kappa^2}, \quad \left\langle \frac{\kappa_1 \kappa_3}{\kappa^2} \right\rangle_\theta = 0, \quad (5.5)$$

with $\kappa^2 = \kappa_2^2 + \kappa_\perp^2$, and all weights mixing a resolved and an unresolved index (e.g. $\kappa_1 \kappa_2 / \kappa^2$) averaging to zero. Substituting (5.5) into (5.3) removes the azimuthal dependence and leaves a two-dimensional integral over $(\kappa_2, \kappa_\perp)$ of the axisymmetric spectrum. Defining the diagonal one-dimensional spectra along the line,

$$E_n^\parallel(\kappa_2) = \int_0^\infty \Phi_{nn}(\kappa_2, \kappa_\perp) 2\pi \kappa_\perp d\kappa_\perp \quad (\text{no sum}), \quad (5.6)$$

the rapid pressure–strain term reduces to a two-dimensional integral over the half-plane $(\kappa_2, \kappa_\perp)$ of the axisymmetric spectrum. The axisymmetric, solenoidal spectrum is fixed by two scalar functions of $(\kappa_2, \kappa_\perp)$ (Batchelor 1946; Chandrasekhar 1950); we denote them $a(\kappa_2, \kappa_\perp)$ and $c(\kappa_2, \kappa_\perp)$, where a is the isotropic-like scalar (the sole survivor when the turbulence is isotropic) and c measures the axisymmetric departure from isotropy. Carrying out the azimuthal average (5.5) and specialising to the plane strain $A = \text{diag}(\frac{1}{2}, -\frac{1}{2}, 0)$, the diagonal rapid pressure–strain components are

$$\Pi_{nn}^{(r)} = \int_0^\infty \int_0^\infty g_n(x) [a(\kappa_2, \kappa_\perp), c(\kappa_2, \kappa_\perp)] 2\pi \kappa_\perp d\kappa_\perp d\kappa_2 \quad (\text{no sum on } n), \quad (5.7)$$

with $x \equiv \kappa_\perp^2 / \kappa^2$, $\kappa^2 = \kappa_2^2 + \kappa_\perp^2$, and the kernels, linear in the two spectral scalars,

$$g_1(x) = a\left(\frac{1}{8} + \frac{3}{4}x - \frac{7}{8}x^2\right) + c\frac{7}{8}x(1-x)^2, \quad (5.8)$$

$$g_2(x) = -\frac{3}{2}x [a(1-x) + c(1-x)^2], \quad (5.9)$$

$$g_3(x) = a\left(-\frac{1}{8} + \frac{3}{4}x - \frac{5}{8}x^2\right) + c\frac{5}{8}x(1-x)^2. \quad (5.10)$$

1126 Equations (5.7)–(5.10) are the central result: *under A1, the rapid pressure–strain term is a*
 1127 *closed, computable functional of the axisymmetric spectral scalars a, c on the resolved half-*
 1128 *plane and the mean strain*, with no reference to the unresolved azimuthal structure beyond
 1129 what A1 fixes. The kernels satisfy two exact constraints that verify the reduction. First,
 1130 energy conservation: pressure–strain redistributes without changing the trace, and indeed
 1131 $g_1(x) + g_2(x) + g_3(x) = 0$ identically for all x and for both scalars, so $\Pi_{11}^{(r)} + \Pi_{22}^{(r)} + \Pi_{33}^{(r)} = 0$.
 1132 Second, the isotropic limit: setting $c = 0$ with a a function of κ alone, the integral of g_3 over
 1133 the half-plane vanishes, so $\Pi_{33}^{(r)} = 0$ —as it must, since the plane strain has no component
 1134 along x_3 and the isotropic rapid term is proportional to S_{ij} (Crow 1968). Both constraints
 1135 hold identically in the derived kernels, confirming the tensor reduction.
 1136

1137 5.4. Residual error and the limit of validity

1138 Assumption A1 is exact for the undistorted inflow and is progressively violated as the
 1139 rapid strain itself renders the spectrum anisotropic about the line. The leading correction
 1140 is governed by the departure-from-axisymmetry tensor $D_{mn}(\boldsymbol{\kappa}) = \Phi_{mn} - \Phi_{mn}^{\text{axi}}$, whose
 1141 contribution to (5.3) is the azimuthal first harmonic that (5.5) discards. Because that
 1142 harmonic integrates to zero against the isotropic part and enters M_{ijkl} only at second
 1143 order in the accumulated anisotropy, the closed form (5.7) is accurate to $O(b^2)$ in the
 1144 Reynolds-stress anisotropy b_{ij} generated by the strain, and exact at $b_{ij} = 0$. The practical

consequence is that (5.7) is most reliable early in the distortion, when the gust first enters the strained region, and degrades as the total strain e accumulates—precisely the regime ordering that the convective mapping of § 4.2 makes physical, since the acoustically relevant upwash is set near the leading-edge plane where the accumulated strain is bounded. We therefore regard (5.7) as a controlled leading-order closure, valid where the gust anisotropy is moderate, rather than as an identity.

5.5. Reduction to the moment closure

The closed form (5.7)–(5.10) operates on the resolved spectrum, whereas the distortion results of § 4.2 were obtained with the algebraic moment closure of § 3.2 (Launder *et al.* 1975). For the two to be consistent, the spectral kernel must reduce to that moment closure in the limit where the moment closure is exact—isotropic turbulence. We show that it does, which both justifies the closure used in the simulations and fixes the prefactor in an explicit convention.

Figure 18 plots the kernels g_1, g_2, g_3 of (5.8)–(5.10) against the angular variable $x = \kappa_2^2/\kappa^2$. Two features are exact and hold for both spectral scalars. The kernels sum to zero at every x ,

$$g_1(x) + g_2(x) + g_3(x) = 0 \quad \forall x, \quad (5.11)$$

so the rapid term conserves turbulent kinetic energy pointwise in wavenumber, not merely in the integral; and each kernel vanishes at $x = 1$, where the wavevector aligns with the line and the projection $\kappa_a \kappa_b / \kappa^2$ degenerates. The signs are physically ordered: $g_1 > 0$ (energy fed to the streamwise component E_1), $g_2 < 0$ (energy removed from the line/upwash component E_2) over the bulk of the angular range, with g_3 changing sign, consistent with the plane strain stretching x_1 and compressing x_2 .

For *isotropic* turbulence the anisotropy scalar vanishes ($c = 0$) and the isotropic-like scalar a depends only on κ . Averaging the kernels over solid angle—equivalently, integrating $g_n(x)$ against the isotropic distribution of x on the sphere—gives

$$\langle g_1 \rangle : \langle g_2 \rangle : \langle g_3 \rangle = +\frac{1}{5} : -\frac{1}{5} : 0, \quad (5.12)$$

in the ratio $+1 : -1 : 0$. This is exactly the structure of the plane strain $S_{ij} = \text{diag}(\frac{1}{2}, -\frac{1}{2}, 0)$: the rapid term is proportional to S_{ij} , with $\Pi_{33}^{(r)} = 0$ because the strain has no component along x_3 . The spectral kernel therefore collapses, under isotropy, to the isotropic-production form of the moment closure,

$$\Pi_{ij}^{(r)} = \frac{4}{5} k_t S_{ij} \quad (\text{isotropic limit}), \quad (5.13)$$

the result of Crow (1968) that underlies the isotropisation-of-production (IP) rapid term in the closure of Launder *et al.* (1975). We note a convention: with the symmetrised definition of the pressure–strain correlation adopted here and the factor of two carried in the Poisson source (2.13), the kernels (5.8)–(5.10) yield the coefficient $\frac{2}{5} k_t$ for the 11-component directly (since $S_{11} = \frac{1}{2}$), equivalent to the standard $\frac{4}{5} k_t S_{ij}$ of (5.13); the ratio (5.12) and the tracelessness (5.11) are independent of this convention.

The consequence is that the algebraic closure used to obtain the distortion of § 4.2 is not an independent assumption but the isotropic-limit reduction of the line-consistent spectral term derived here. The spectral form (5.7) carries, in addition, the angular (x -dependent) and anisotropy (c -scalar) information that the moment closure discards; the corrections it retains are precisely those of order b^2 in the strain-generated anisotropy identified in § 5.4. The moment closure and the spectral closure thus agree to leading order and depart in

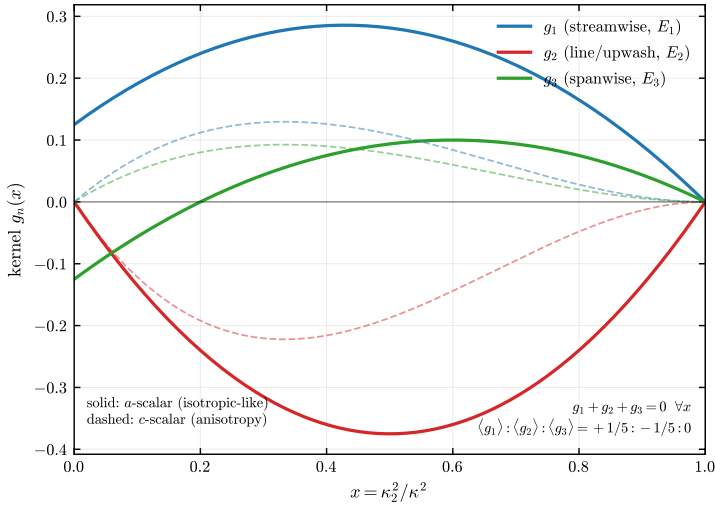


Figure 18. Rapid pressure–strain closure kernels g_1, g_2, g_3 (eqs. (5.8)–(5.10)) against the angular variable $x = \kappa_2^2/\kappa^2$. Solid curves: coefficient of the isotropic-like spectral scalar a ; faint dashed curves: coefficient of the anisotropy scalar c . The kernels sum to zero at every x (energy conservation, eq. (5.11)) and vanish at $x = 1$. Averaged over an isotropic spectrum they give the ratio $+1/5 : -1/5 : 0$ (eq. (5.12)), recovering the isotropisation-of-production form $\Pi_{ij}^{(r)} = \frac{4}{5} k_t S_{ij}$ (Crow 1968) that underlies the moment closure used in § 4.2.

1177 a controlled way as the gust anisotropy grows, which is the regime ordering the present
 1178 application requires.
 1179

1180 5.6. Consistency with the eddy redistribution: avoiding double counting

1181 A closed rapid term is necessary but not sufficient: it must not duplicate redistribution
 1182 already effected by the eddy events. The pressure–strain correlation decomposes canon-
 1183 ically into rapid and slow parts, $\Pi_{ij} = \Pi_{ij}^{(r)} + \Pi_{ij}^{(s)}$, corresponding to the two source
 1184 terms in (5.1). The slow part $\Pi_{ij}^{(s)}$ is the return-to-isotropy mechanism, quadratic in the
 1185 fluctuations and independent of the mean gradient; the eddy events in ODT, which act on
 1186 the fluctuating field without reference to A_{kl} , are the model’s representation of exactly this
 1187 slow redistribution. The rapid part $\Pi_{ij}^{(r)}$, proportional to A_{kl} and linear in the fluctuations,
 1188 is by construction absent from the eddy mechanism, which carries no dependence on the
 1189 mean strain. The two therefore occupy disjoint terms of the canonical decomposition, and
 1190 adding (5.7) as an explicit mean-strain-driven source alongside the eddies does not double-
 1191 count, provided the eddy events are calibrated to reproduce the slow return-to-isotropy
 1192 in the absence of mean strain—a condition that the no-strain baseline of § 4.2.3 verifies
 1193 directly, since there the centroid relaxes under the eddies alone with no spurious rapid
 1194 contribution. This argument establishes consistency at the level of the decomposition; the
 1195 no-strain baseline of § 4.2.3 provides the direct numerical confirmation, the centroid there
 1196 relaxing under the eddy mechanism alone.

1197

1198 5.7. Summary

1199 The rapid pressure–strain term, exactly a nonlocal three-dimensional integral (5.3) of the
 1200 full spectrum, collapses under axisymmetry of the undistorted inflow about the resolved

direction (Assumption A1) to the closed, single-line functional (5.7) of the resolved spectra and the mean strain. The collapse is exact for the isotropic inflow and controlled to $O(b^2)$ in the strain-generated anisotropy, with the error growing as the accumulated strain renders the spectrum anisotropic about the line. The resulting rapid source is consistent with—and disjoint from—the slow redistribution carried by the eddy events, so the strain-coupled model is free of double counting at the level of the rapid/slow decomposition. The closure therefore holds on a single line within a stated and physically grounded limit, rather than universally; characterising its breakdown at large accumulated strain, where the gust anisotropy is no longer moderate, remains for future work.

6. Conclusions

Standard one-dimensional turbulence carries no mechanism for the action of an imposed mean strain on the fluctuating field: its pressure treatment models only the slow, turbulence-driven return to isotropy, not the rapid, mean-strain-driven redistribution that governs the distortion of turbulence as it approaches a leading edge. This paper has presented a strain-coupled formulation that supplies the missing physics while preserving the structural foundations of the model. The mean-strain production enters as a continuous, pointwise forcing of the line velocity, which we proved preserves the measure-preservation, continuity, scale-locality and kernel identities of the triplet-map mechanism (Proposition 1); the rapid pressure–strain enters as an energy-conserving redistribution operator, reconstructed at each step from the running Reynolds stress through a Lyapunov equation and constructed to reproduce the homogeneous rapid-distortion evolution of the component energies exactly under an explicit spectral closure (Proposition 2); and the kinematic compression of scales is carried by a consistent dilatation of the ODT domain.

Two questions framed the contribution. The first—whether the distorted spectrum departs measurably from the rigid translation that linear rapid-distortion theory predicts—is answered in the affirmative by § 4.2: at the Reynolds number of the target inflow the strain-coupled model produces a broadband distorted spectrum whose energy-weighted centroid migrates to ≈ 2.4 by a total strain of four, far below the rigid-translation value of 7.4, a difference of spectral shape that a frozen, linearly strained input cannot represent. The second—whether the rapid pressure–strain, exactly a nonlocal three-dimensional integral of the full spectrum, admits a consistent representation on a single one-dimensional line—is answered in § 5: under the single, physically grounded assumption that the undistorted inflow is axisymmetric about the line, the integral collapses to a closed, computable kernel of the resolved spectra and the mean strain, exact in the isotropic limit and controlled to second order in the strain-generated anisotropy, and disjoint from the slow redistribution carried by the eddy events. The closure thus holds on a single line within a stated limit rather than universally, and that limit is the regime the leading-edge problem occupies.

The formulation was verified against analytical rapid-distortion theory, which it reproduces exactly at the onset of straining with no adjustable constant, and validated against two external benchmarks: the strained-turbulence experiment of [Chen *et al.* \(2006\)](#), whose rapid-distortion limit the model reproduces exactly and whose full-model spectral anisotropy it captures qualitatively, and the direct numerical simulations of [Lee & Reynolds \(1985\)](#), against which the model reproduces the production-governed Reynolds-stress anisotropy quantitatively. Where the model departs from the latter—in the unstrained-direction anisotropy under plane strain—the departure was shown to lie outside the reach of rapid-distortion theory and of the entire second-moment-closure family, so that the model performs at the level of the best available closure on this benchmark.

The output of the model is the distorted, anisotropic upwash spectrum at the leading-edge plane, delivered with full scale resolution at a cost—some 300 core-hours for the resolved $N = 200$ ensemble, roughly two orders of magnitude below an equivalent three-dimensional simulation—that makes parametric study across turbulence intensity, length scale and Reynolds number practical. Supplying that spectrum to the leading-edge response of Amiet (1975), in place of the prescribed isotropic input of the classical prediction chain, is the natural application of the present development and is the subject of a companion paper; the remaining ingredients of a complete leading-edge-noise framework—the kinematic surface blocking that suppresses the wall-normal component near the surface, and the spanwise coherence of the distorted field—are the principal directions for future work.

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8. Declaration of the use of AI tools

The author used a large language model, Apertus-8B-Instruct-2509 (Swiss AI Initiative 2025) (Swiss AI Initiative; hosted on the Blablador service of the Jülich Supercomputing Centre, Helmholtz Association; <https://huggingface.co/swiss-ai/Apertus-8B-Instruct-2509>; accessed July 2026), for English-language grammar and phrasing checks of the author’s own draft text. The tool was not used to generate scientific content, to produce or analyse data, or to generate figures; all derivations, numerical results, figures and interpretations are the author’s own. Accountability for the final text rests with the author, who has checked the manuscript, and in particular the references, for any unintended consequences of this use.

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